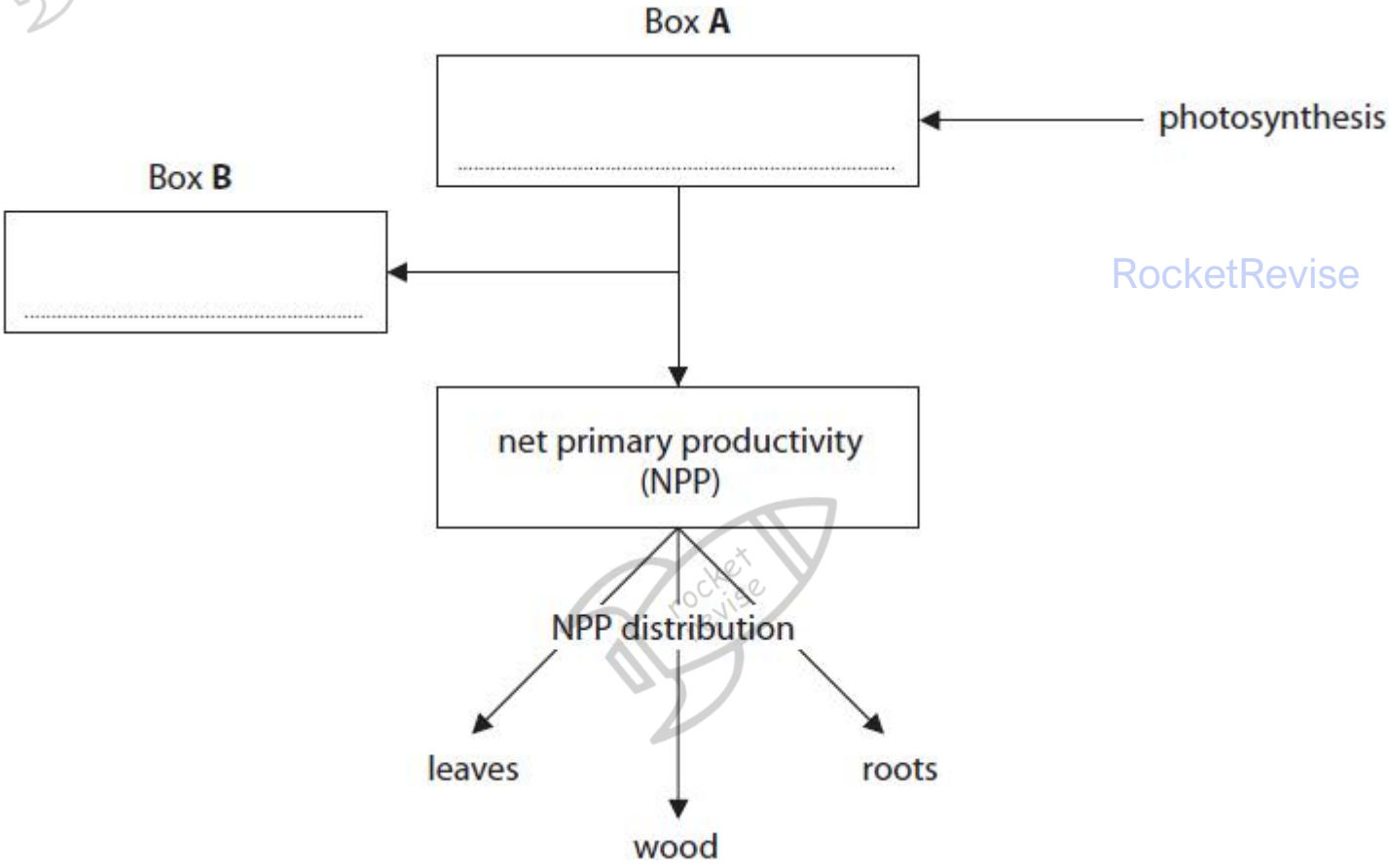
www.RocketRevise.com

Questions

Q1.

The diagram shows some processes linking photosynthesis to productivity and the distribution of biomass in the trees in a forest.



(a) Complete the diagram by writing the appropriate word or words in box **A** and in box **B**.

(1)

(b) In a healthy forest ecosystem, the NPP should be distributed equally to the leaves, wood and roots.

(i) State the meaning of the term **ecosystem**.

(1)

.....

.....

.....

(ii) Which of the following are the units for NPP?

(1)

- ☒ **A** kg m^{-1} yr^{-1}
- ☒ **B** kg m^{-2} yr^{-2}
- ☒ **C** kJ m^{-1} yr^{-1}
- ☒ **D** kJ m^{-2} yr^{-1}

(iii) In this forest, the total NPP was 11 700 units.
The leaves contained 34% of the NPP and the wood contained 39% of the NPP.
Calculate the NPP content in the roots.

(2)

www.RocketRevise.com

www.RocketRevise.com

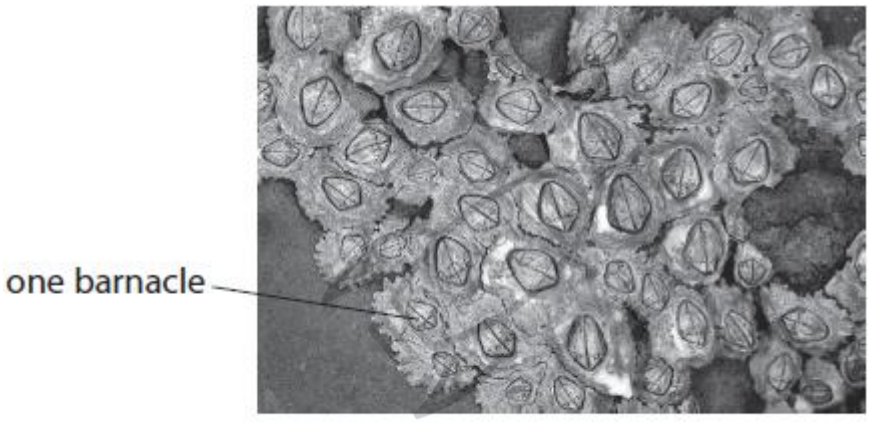
Answer ..... units

(Total for question = 5 marks)

Q2.

The photograph shows barnacles attached to a rock on a seashore.

RocketRevise



© FLPA/Alamy Stock Photo

The list gives some facts about barnacles:

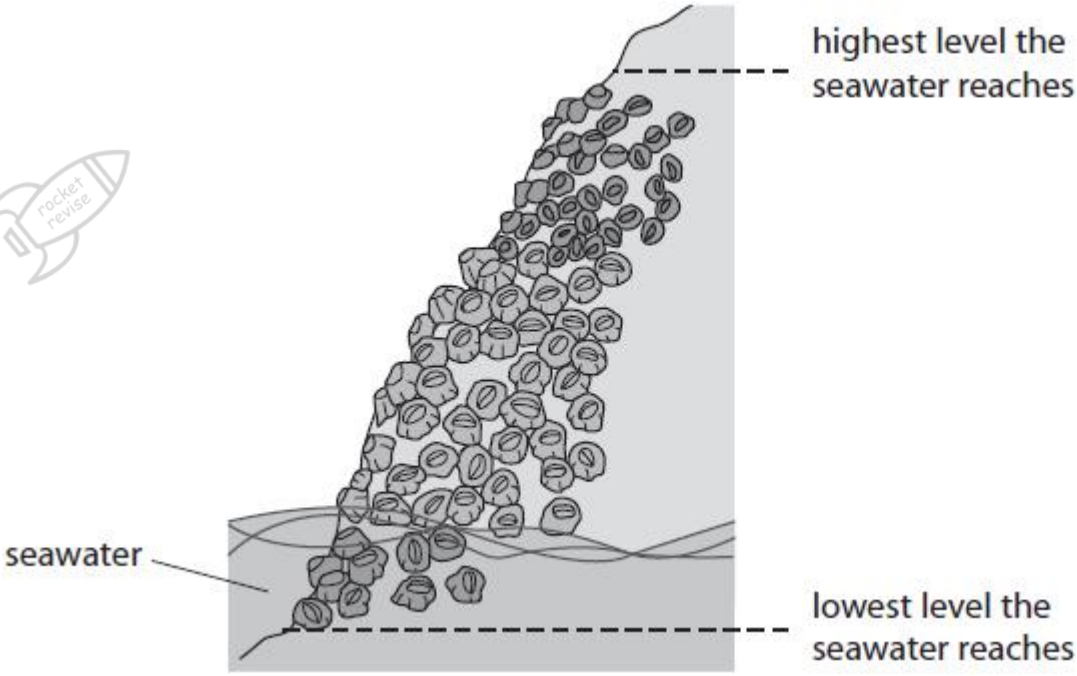
- barnacles are marine animals that live inside their shells
- the shells are permanently attached to a solid surface
- barnacles feed on food particles that they filter from the water
- barnacles reproduce sexually and produce larvae that swim in the seawater
- larvae settle on suitable surfaces such as rocks.

(a) The diagram shows the distribution of two types of barnacle, *Chthamalus* and *Balanus*, on a rock.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

These two types of barnacle occupy different niches.
Explain how the distribution and abundance of these two types of barnacle on this rock show that these barnacles occupy different niches.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

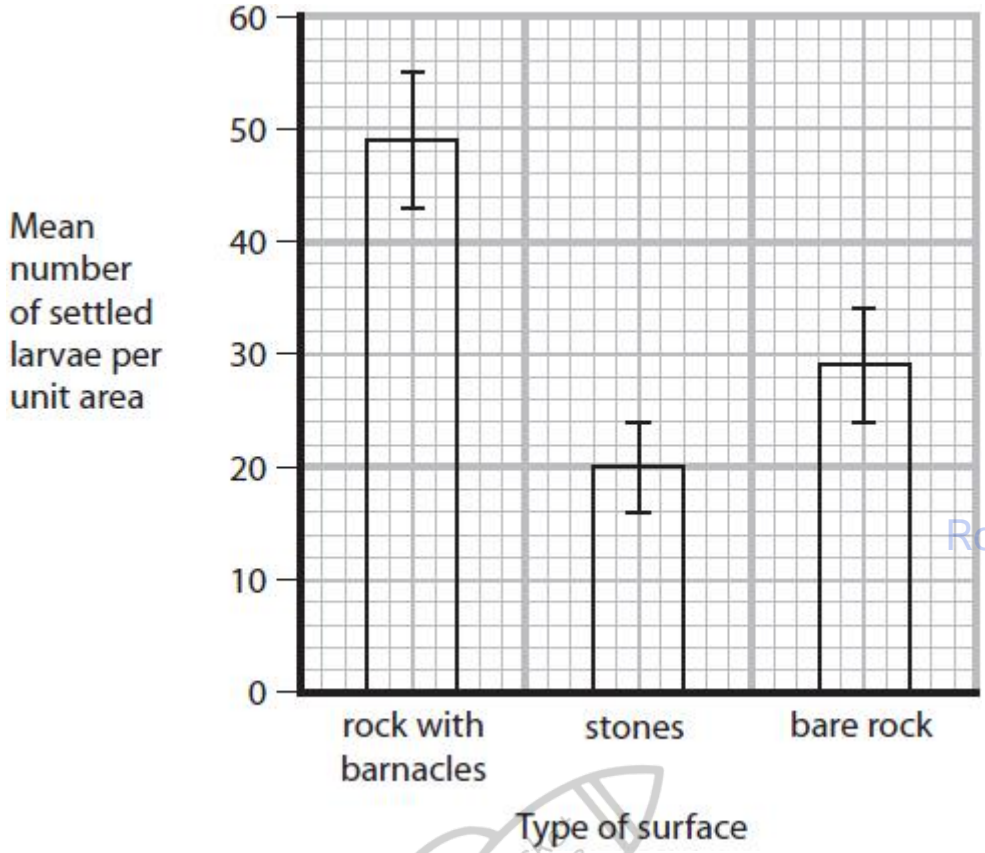
.....

(b) The types of surface that the larvae settled on were studied.
The graph shows the results of this study.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(i) Comment on the reliability of the conclusions that can be drawn from this study.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

(ii) Describe how this study could be carried out in a laboratory.

(4)

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 11 marks)

Q3.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Mosses are species of plants that have genomes that enable them to survive in their habitats.

(a) A group of students investigated the mean number of plant species in four habitats.

The habitats that they investigated were:

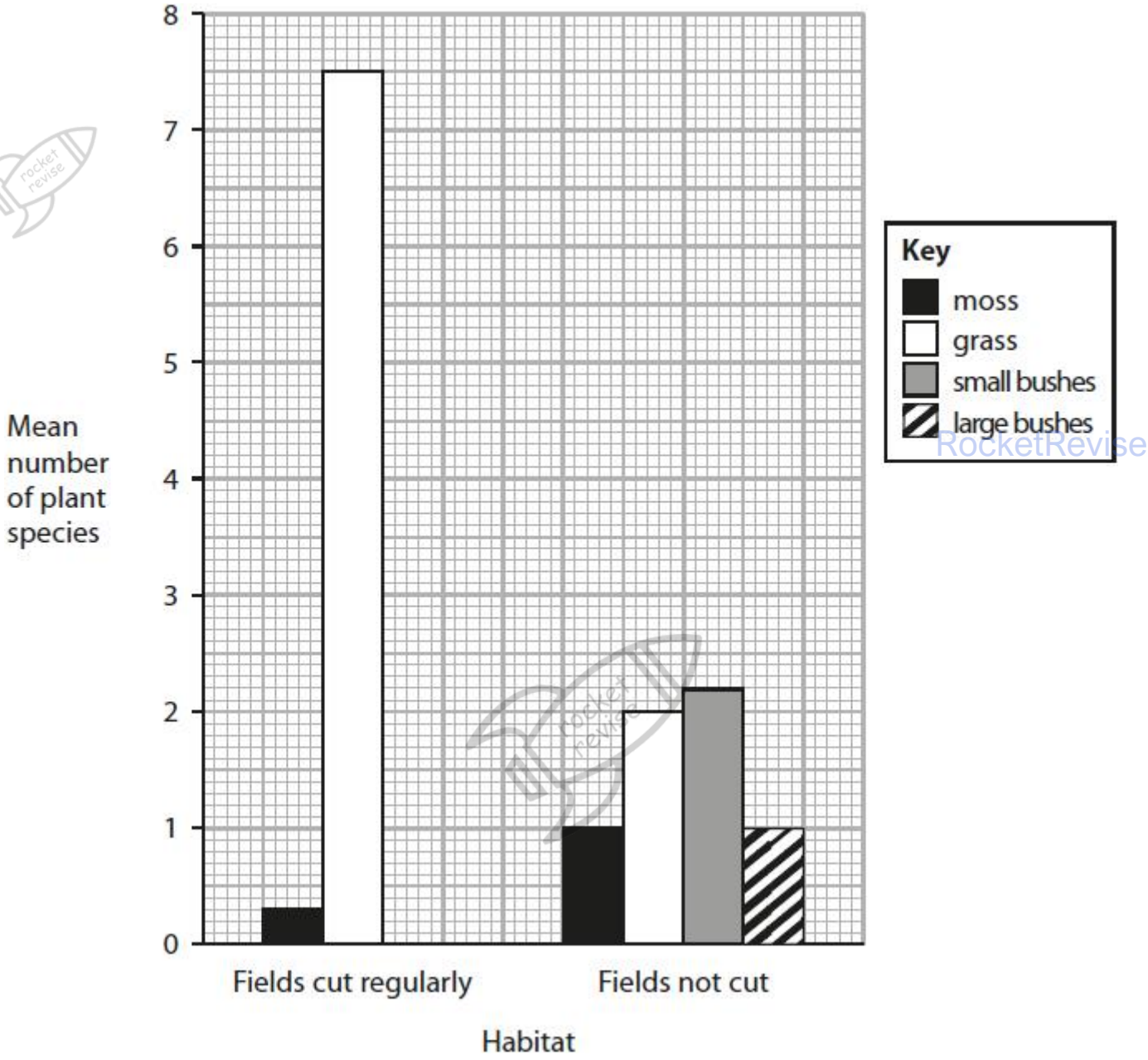
- fields that were cut regularly
- fields that were not cut
- the middle of woods
- the edge of woods.

The graph shows some of the results from this investigation.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



(i) Which row of the table shows the type of factor that affected the plant species in two of these habitats?

(1)

	Cutting the field regularly	Competition with trees for water
☐ A	abiotic factor	abiotic factor
☐ B	abiotic factor	biotic factor
☐ C	biotic factor	abiotic factor
☐ D	biotic factor	biotic factor

(ii) What is the percentage increase in the number of species of moss between the fields cut regularly and the fields not cut?

(1)

☐ **A** 23.3
☐ **B** 70.0
☐ **C** 233.3

www.RocketRevise.com

www.RocketRevise.com

☒ **D** 333.3

(iii) Suggest reasons for the differences in the mean number of plant species in fields cut regularly and in fields not cut.

(2)

.....

.....

.....

.....

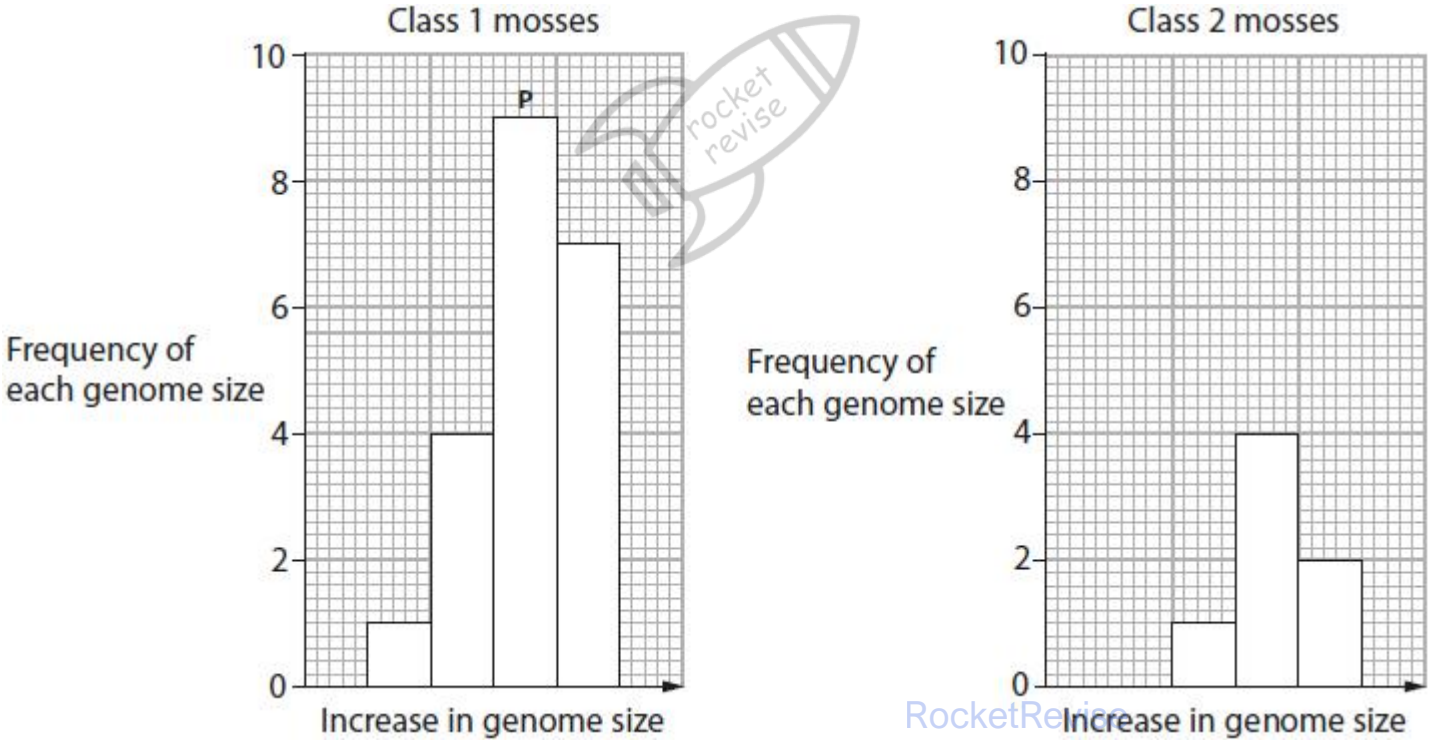
.....

.....

RocketRevise

(b) In another investigation, the genome size of 30 species of moss was determined. Each species of moss belonged to one of two classes of moss.

The graphs show the frequency of each genome size in the class 1 mosses and some of the class 2 mosses.



(i) Which does **P** represent?

(1)

☒ **A** the maximum genome size on a bar chart
☒ **B** the maximum genome size on a histogram
☒ **C** mode genome size on a bar chart
☒ **D** mode genome size on a histogram

(ii) One genome size is missing from the graph for class 2 mosses.
Calculate the frequency of this genome size.

(1)

www.RocketRevise.com

www.RocketRevise.com

Answer .....

(iii) The mean value for the genome size of class 1 mosses was 0.449 a.u. and the mean value for class 2 mosses was 0.920 a.u.

Calculate the ratio of the genome size of class 1 mosses to the genome size of class 2 mosses.

(1)

RocketRevise

Answer .....

(iv) It was suggested that the chromosomes in class 2 mosses were found in pairs.

Give the evidence that supports this suggestion.

(1)

.....

.....

.....

(Total for question = 8 marks)

Q4.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a polar bear.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(Source: © Burcu Gultekin / Alamy Stock Photo)

Polar bears are adapted to live in the Arctic Circle.

Polar bears have a layer of fat beneath their skin. This fat insulates them from the cold and can be used to supply the polar bear with water (metabolic water).

They have a thick fur coat which also insulates them from the cold.

Polar bears feed on seals. The polar bear sits on the ice and waits for a seal to come to the surface of the water to breathe.

(a) Which row of the table describes the types of adaptation of polar bears?

(1)

	Anatomical	Behavioural	Physiological
☐ **A**	produces metabolic water	hunting by sitting on ice	thick fur coat
☐ **B**	hunting by sitting on ice	thick fur coat	produces metabolic water
☐ **C**	thick fur coat	produces metabolic water	hunting by sitting on ice
☐ **D**	thick fur coat	hunting by sitting on ice	produces metabolic water

(b) What is the niche of a polar bear?

(1)

- ☐ **A** produces metabolic water
- ☐ **B** the Arctic Circle
- ☐ **C** controls the population size of seals
- ☐ **D** uses the layer of fat beneath the skin

(c) Suggest why the polar bear needs to rely on the store of fat to provide it with water.

(2)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(d) Climate change is resulting in habitat fragmentation. Habitat fragmentation causes groups of polar bears to become separated from one another.

Explain the effect that habitat fragmentation could have on the genetic diversity of polar bears.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 7 marks)

Q5.

The photograph shows two zebras.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



(Source: Caroline Wilcox)

RocketRevise

The stripes of zebras are thought to reduce attack by biting flies.

Biting flies cause stress and disease to cattle, reducing food production and causing financial losses for farmers.

(a) Farmers use chemicals to kill the flies.

Flies often evolve resistance to a new chemical within about 10 years.

(i) Explain how flies evolve resistance to new chemicals.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(ii) Suggest why the evolution of resistance to these chemicals occurs so quickly in flies.

(1)

.....

.....

.....

*(b) An investigation was carried out to see if the presence of stripes on cattle reduced the number of biting flies on their bodies.

Three groups of black cattle were compared:

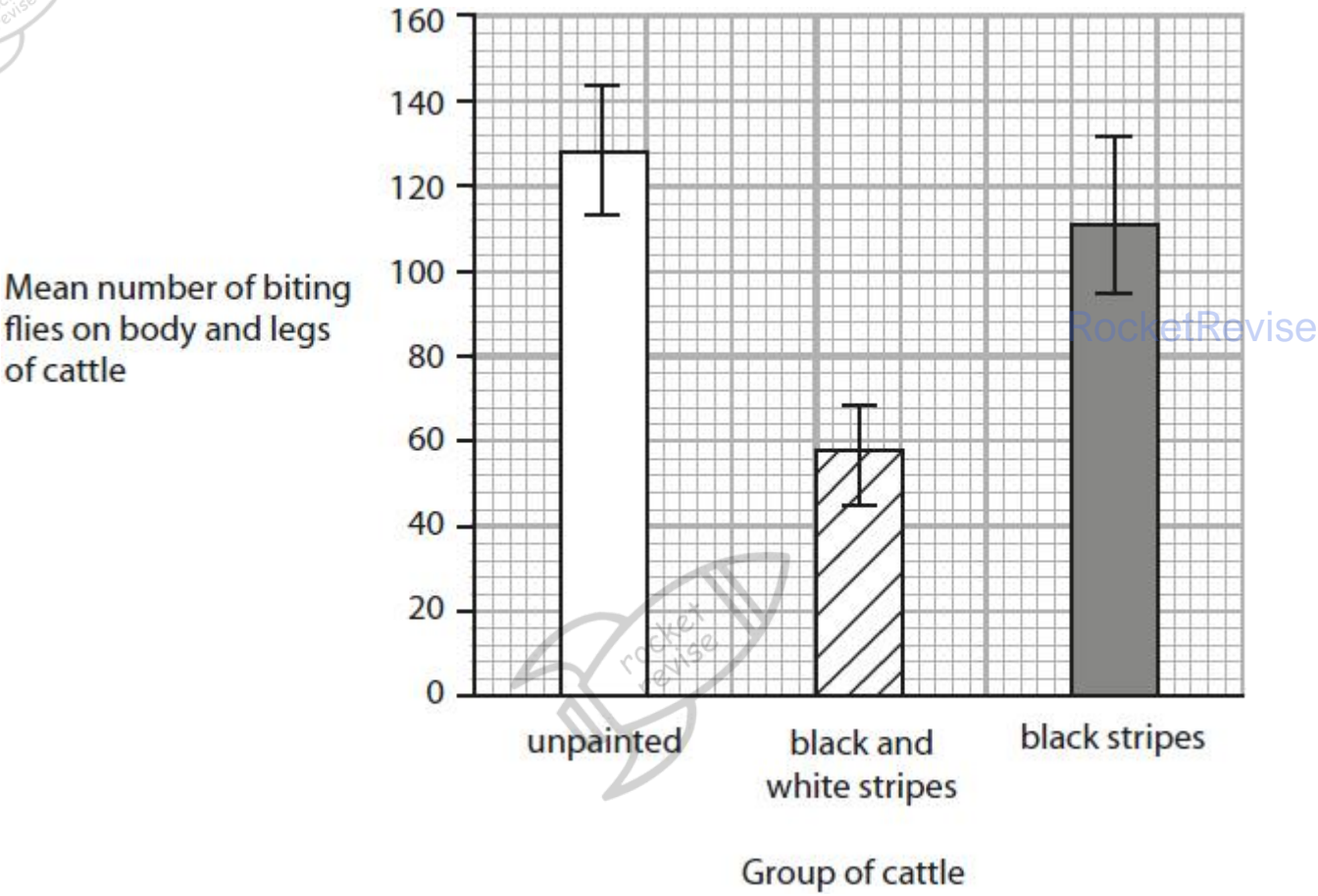
- unpainted cattle

www.RocketRevise.com

www.RocketRevise.com

- cattle painted with black and white stripes
- cattle painted with black stripes only.

Cattle repel flies by flicking their tails, stamping their feet and skin twitching.
The frequencies of these fly-repelling behaviours were recorded.
The graph and tables show some of the results of this investigation.



RocketRevise
RocketRevise
www.RocketRevise.com

www.RocketRevise.com

Group of cattle	Number of flies on the body	Number of flies on the legs
unpainted	662	1309
black and white stripes	231	710
black stripes	677	1030

Table 1

Group of cattle	Mean frequencies of fly-repelling behaviours / number per hour		
	Flicking their tails	Stamping their feet	Skin twitching
unpainted	54	16	5
black and white stripes	42	10	8
black stripes	55	16	5

Table 2

Determine the effectiveness of painting stripes on cattle in reducing the number of biting flies on them.
Use the information in the graph and both tables to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

(Total for question = 10 marks)

Q6.

Reforestation is an example of effective management of conflicts between human needs and conservation.

Organisations have put together a list of rules for responsible reforestation. These rules aim to improve absorbing and storing atmospheric carbon dioxide, biodiversity and human livelihoods.

Four of the rules are:

1. Protect existing forests first
2. Work together with local people
3. Select appropriate areas for reforestation
4. Select species to maximise biodiversity.

(a) Explain the importance of protecting existing forests first (rule 1).

(2)

RocketRevise

.....

.....

.....

.....

.....

.....

(b) Suggest why it is important that organisations involved in reforestation work with the local people (rule 2).

(2)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(c) Explain **two** reasons why previously forested areas should be selected for reforestation instead of other wild areas (rule 3).

(4)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(d) Explain why a mixture of native species, including endangered and genetically diverse species, should be introduced (rule 4).

(4)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

(Total for question = 12 marks)

Q7.

RocketRevise

As the human population in the world has increased, the demand for food has increased.

The production of food has an impact on the environment, unless it can be done in a sustainable way.

(a) (i) State the meaning of the term **population**, as it has been used in this context.

(1)

.....

.....

.....

(ii) State the meaning of the term **sustainable**, as it has been used in this context.

(1)

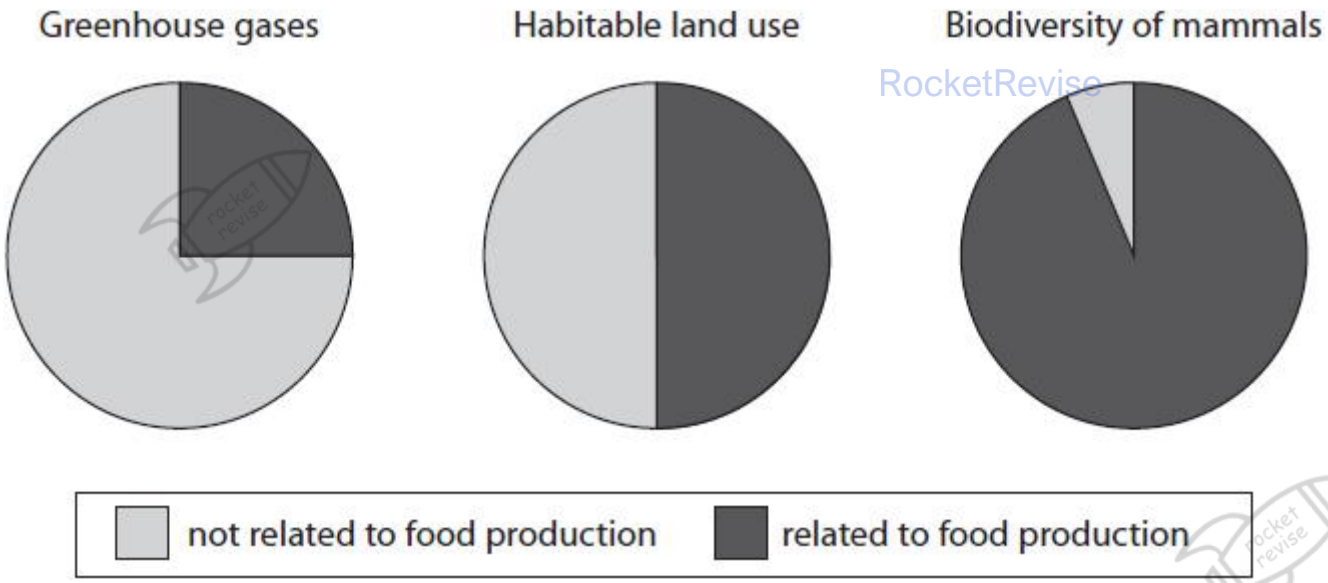
.....

.....

.....

.....

(b) The diagram shows the impact of food production on three environmental factors.



www.RocketRevise.com

www.RocketRevise.com

(i) Explain how food production could contribute to the greenhouse effect.

(3)

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

(ii) Land occupies 149 million km^2 of the surface of the Earth.
Habitable land is 71% of this area and some of it is used for food production.
Calculate the area of the surface of the Earth used in food production.
Express your answer in standard form.

(3)

Answer ..... km^2

(iii) Estimate the effect of food production on the ratio of biodiversity of mammals.

(1)

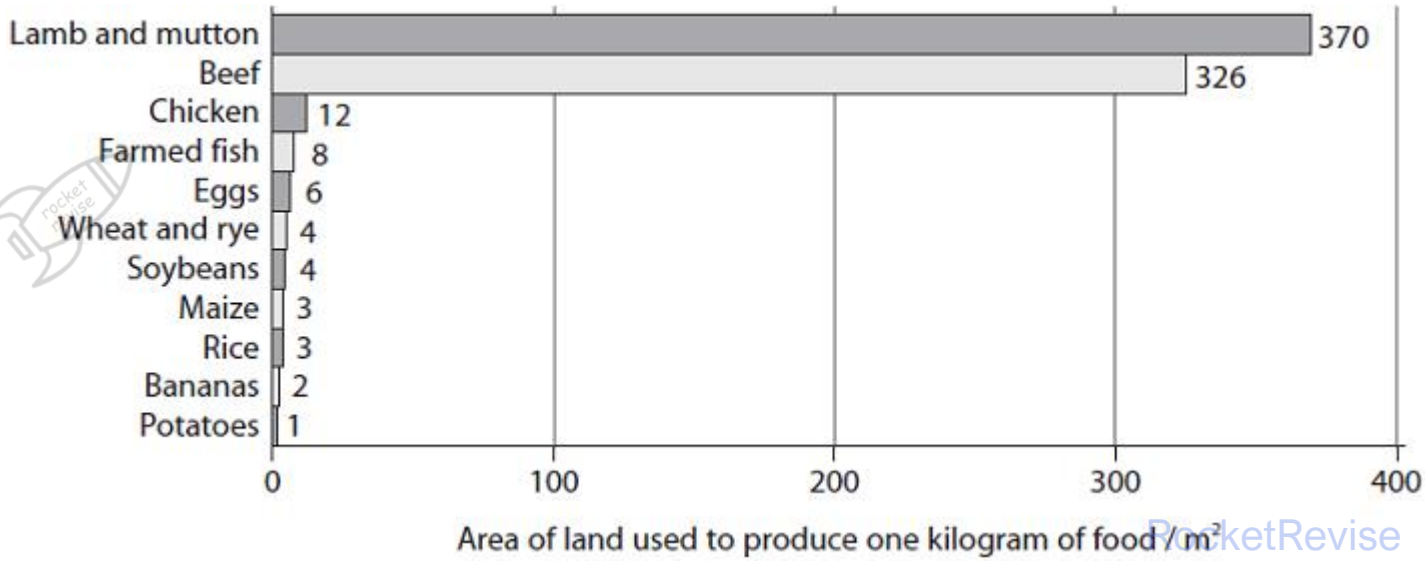
RocketRevise

Answer .....

*(c) The diagram shows the area of land used to produce one kilogram of some food products.

www.RocketRevise.com

www.RocketRevise.com



Explain how food production could be made more sustainable.
Use the information in the diagram and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 15 marks)

www.RocketRevise.com

www.RocketRevise.com

Q8.

The photograph shows a moose.

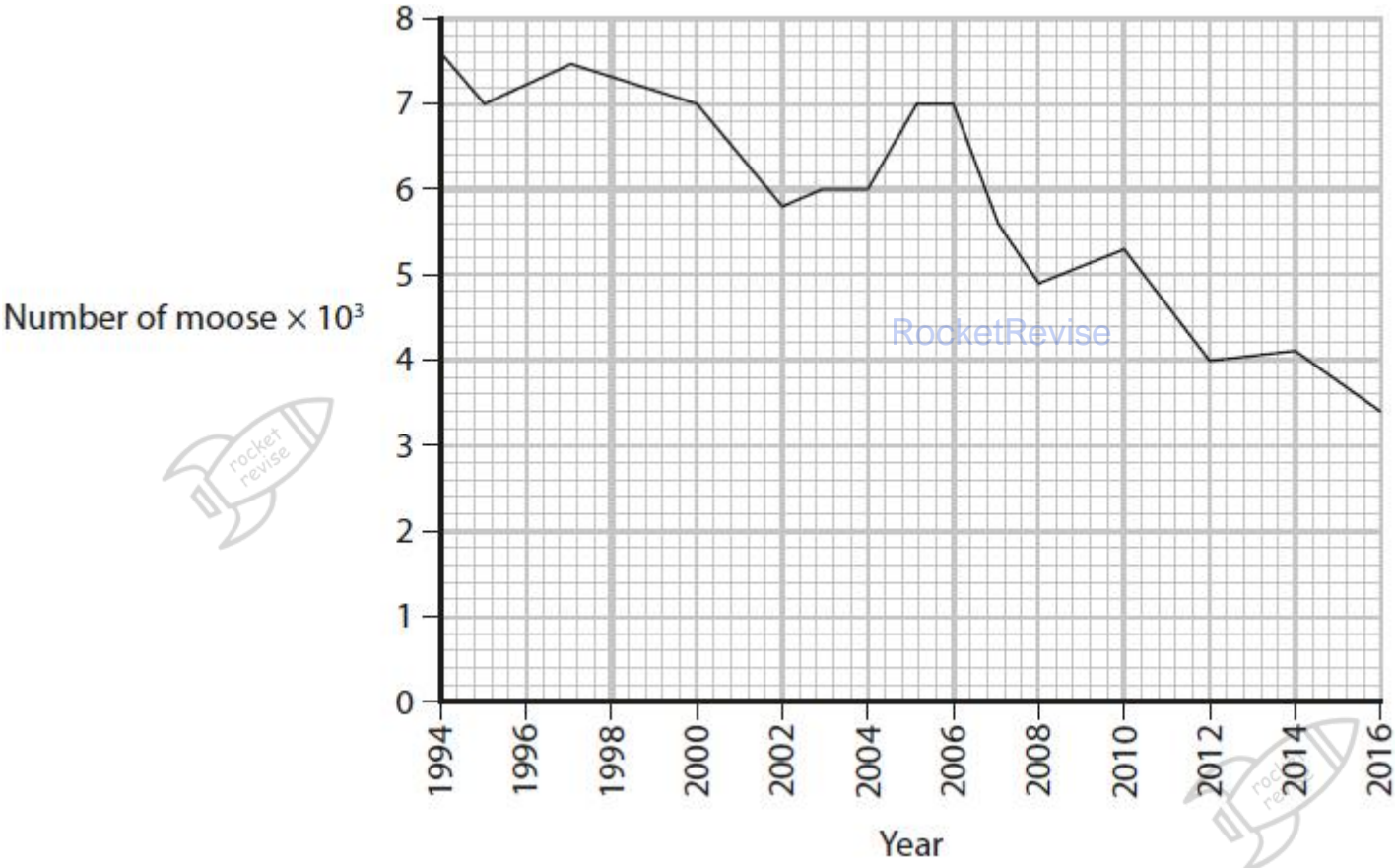


(Source: © blickwinkel/Alamy Stock Photo)

Moose are large mammals that eat grass and the shoots of plants.

The number of moose is declining in parts of America.

(a) The graph shows the number of moose in one part of America from 1994 to 2016.



www.RocketRevise.com

www.RocketRevise.com

(i) In 2016, the moose were not counted. The value shown on the graph is an estimate.
Describe how these data were used to estimate the number of moose in 2016.

(1)

.....

.....

.....

(ii) Calculate the percentage decrease in the number of moose from 1994 to 2016.

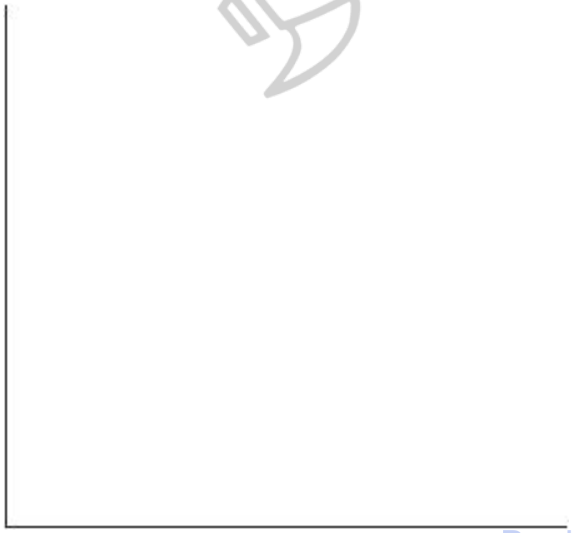
(2)

RocketRevise

Answer ..... %

(iii) It has been suggested that global warming is responsible for the decrease in the number of moose from 1994 to 2016.
Sketch a graph to show this relationship.

(3)



RocketRevise

(b) Ticks are parasites that live in the fur of moose.

Ticks are thought to be responsible for the decline in the number of moose.
The table shows the frequency distribution of ticks in a population of moose.

www.RocketRevise.com

www.RocketRevise.com

Number of ticks	Number of moose
1 to 9999	32
10000 to 19999	46
20000 to 29999	41
30000 to 39999	32
40000 to 49999	22
50000 to 59999	15
60000 to 69000	9
70000 to 79999	4
80000 to 89999	4
90000 to 99999	3
100000 or more	6

RocketRevise

(i) Calculate the percentage of moose that have 50 000 or more ticks on their bodies.

(2)

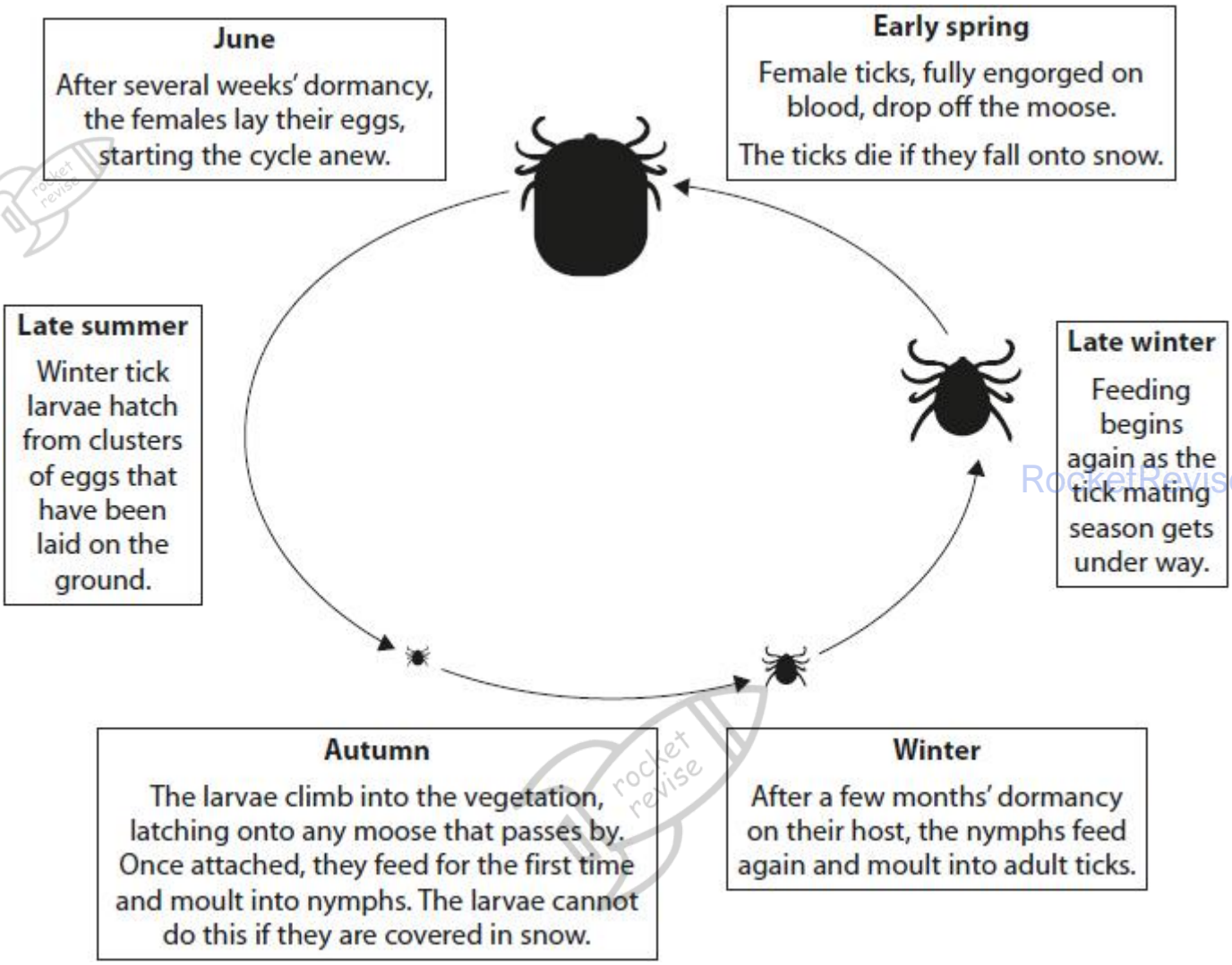
Answer ..... %

*(ii) The diagram shows the life cycle of the tick.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



The ticks feed on the blood of the moose. One tick can remove 200 to 600 times its body mass. The ticks irritate the moose. The moose will scratch against trees causing large clumps of fur to fall off.

Explain how global warming might affect the life cycle of ticks and result in the decline in the number of moose.

Use all the information in the question to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 14 marks)

Q9.

The eruption of a volcano can devastate the surrounding countryside and cause climate change.

(a) The photograph shows the devastation that can result from a volcanic eruption.



(Source: © Zoonar GmbH / Alamy Stock Photo)

Succession occurs in the devastated areas.
Pioneer species are the first to colonise these areas.
Several stages follow, which may result in a climax community.

(i) Describe **three** characteristics of pioneer species.

(3)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

(ii) Explain why succession occurs in stages.

(2)

RocketRevise

.....

.....

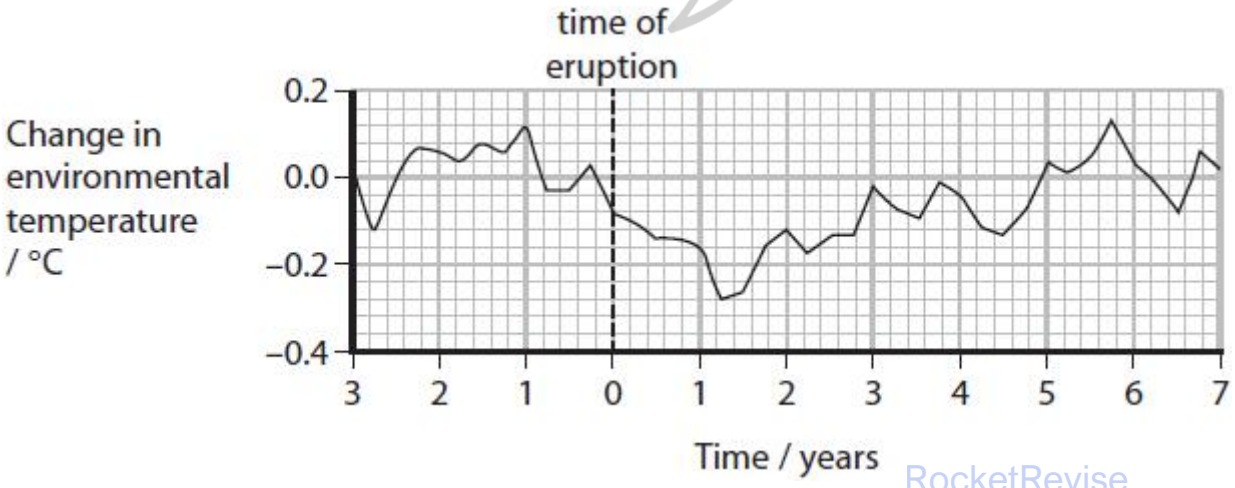
.....

.....

.....

.....

(b) (i) The graph shows the changes in environmental temperature before and after a volcanic eruption.



RocketRevise

Determine the effect of the volcanic eruption on environmental temperature.
Use the information in the graph to support your answer.

(2)

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

(ii) When this volcano erupted, sulfur dioxide and particles of ash were released into the atmosphere.
Sulfur dioxide and particles of ash cause less light to pass through the atmosphere.
Explain the results shown in the graph.

(4)

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

.....

.....

(iii) The eruption of one volcano resulted in 5×10^7 metric tons of carbon dioxide entering the atmosphere in one year.
The mass of carbon dioxide released into the atmosphere in one year by anthropogenic activities was 3.5×10^{10} metric tons.
Calculate how many volcanic eruptions would need to occur, in one year, to produce as much carbon dioxide as anthropogenic activities.

(1)

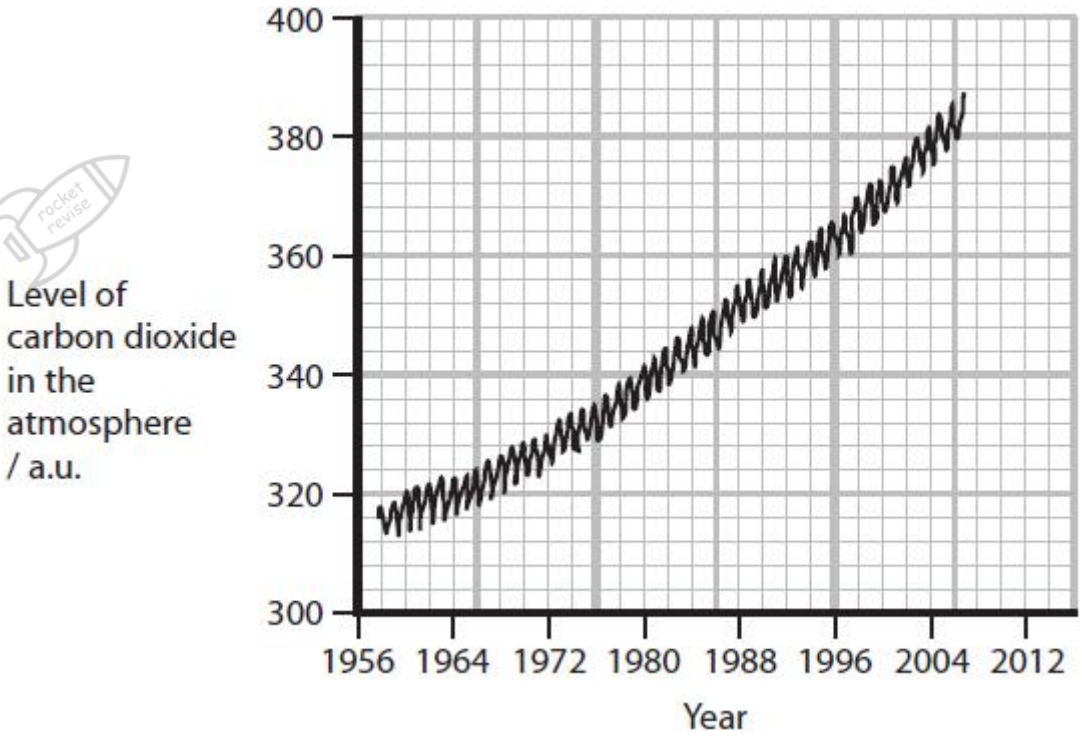
RocketRevise

Answer .....

(iv) The graph shows the levels of carbon dioxide in the atmosphere in one region of the world from 1958 to 2004.
During this period there were volcanic eruptions in 1963, 1980, 1982 and 1991, in this region.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

Discuss the possible effects of volcanic eruptions on climate change.
Use all the information in part (b) to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 15 marks)

Q10.

The photograph shows a wood frog.

www.RocketRevise.com

www.RocketRevise.com



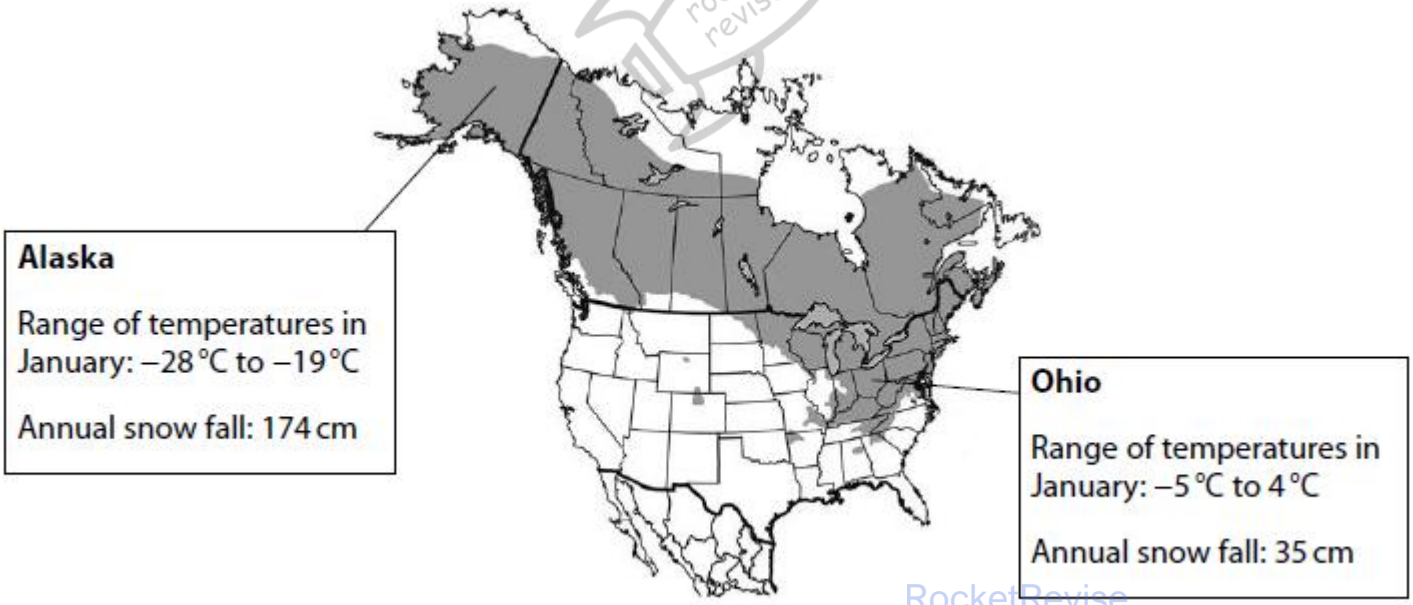
RocketRevise

Source: http://www.borealforest.org/reptiles/wood_frog.htm

Wood frogs are found throughout North America.

The shaded areas on the map shows the distribution of wood frogs in North America.

Information is also given about the climate in two areas, Alaska and Ohio.



RocketRevise

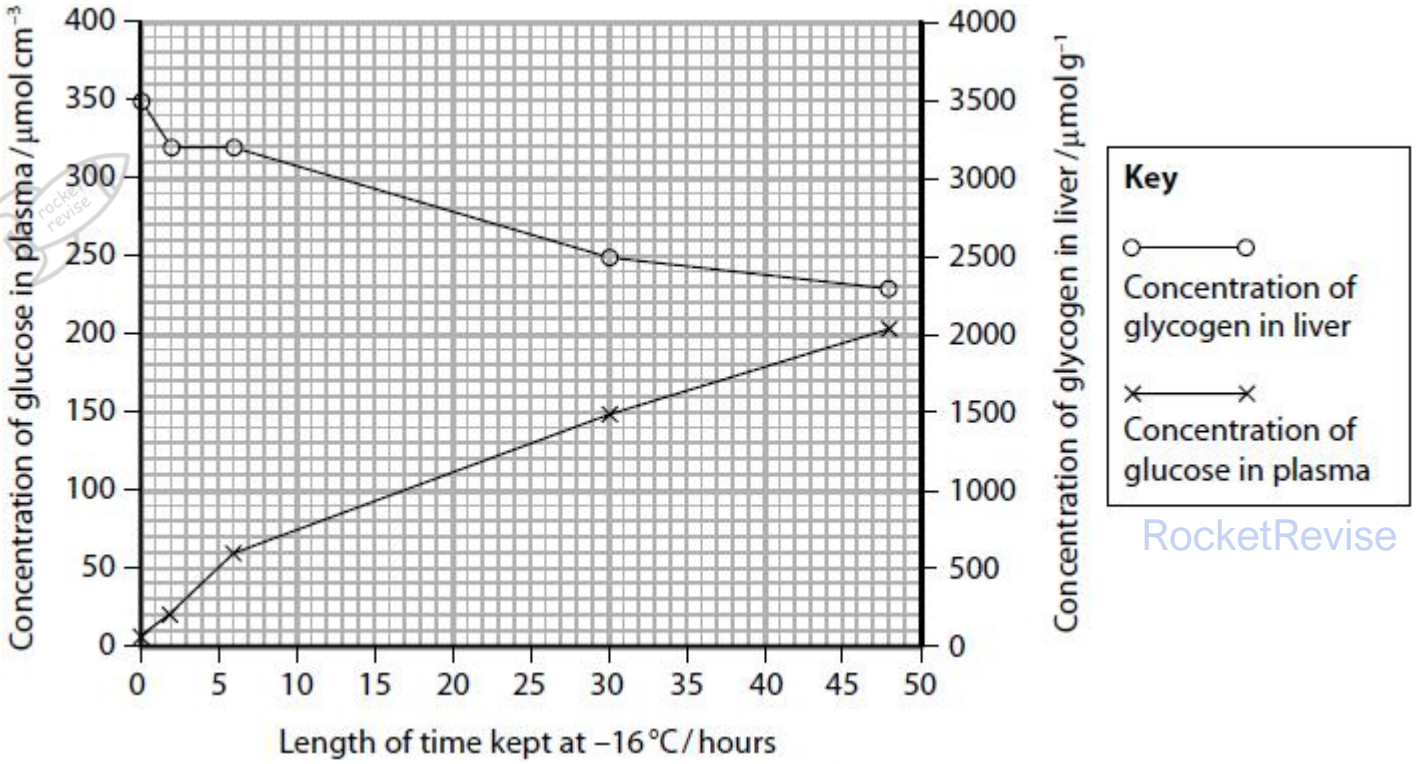
Source: https://answersingenesis.org/natural-selection/adaptation/the-secret-lives-of-frozen-frogs/

(a) Scientists kept a wood frog from Alaska at −16 °C. The scientists measured the concentration of glycogen in the liver and the concentration of glucose in the plasma over a period of 48 hours.

The graph shows the results.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(i) How many of the following statements describe glycogen?

1. The monomer is *β* glucose
2. It is made from two different types of polymer
3. There are 1,4 and 1,6 glycosidic bonds
4. It is insoluble

(1)

- ☒ **A** 1
- ☒ **B** 2
- ☒ **C** 3
- ☒ **D** 4

(ii) Explain the changes in the concentration of glycogen and glucose.
Use the information in the graph to support your answer.

(2)

.....

.....

.....

RocketRevise

.....

.....

.....

(b) The scientists found that wood frogs from Alaska had higher concentrations of glucose in their plasma than wood frogs from Ohio.

The scientists also found that the wood frogs from Alaska had higher concentrations of other solutes, such as urea, in their plasma.
The table shows the mean concentration of urea in the plasma of these frogs. The table also shows the standard deviations.

www.RocketRevise.com

www.RocketRevise.com

Type of wood frogs	Mean concentration of urea in plasma / μmol cm^{-3}	Standard deviation
from Alaska	106	10
from Ohio	28	5

(i) Calculate the percentage difference in the mean concentration of urea in the plasma of the wood frogs from Alaska compared with the wood frogs from Ohio.

(1)

RocketRevise

Answer ..... %

(ii) Explain why the data for the wood frogs from Alaska are more reliable than the data for the wood frogs from Ohio.

(2)

.....

.....

.....

.....

.....

.....

(iii) Explain why a high concentration of solutes in the plasma could protect the wood frogs from Alaska in very cold temperatures.

(2)

.....

.....

.....

.....

.....

.....

(c) Explain how these wood frogs evolved to occupy different niches in North America.

(5)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

(Total for question = 13 marks)

Q11.

Scientists studied the distribution of biomass in organisms on Earth.

(a) State the meaning of the term **biomass**.

(1)

.....

.....

.....

(b) Voronoi diagrams were used to present some of the data.

RocketRevise

Two Voronoi diagrams are shown.
Diagram A shows the biomasses of groups of organisms.
Diagram B shows the biomasses of the organisms in the animals group.
The biomass is given in gigatons of carbon (Gt C), where 1 Gt C = 10^{15} g of carbon.

www.RocketRevise.com

www.RocketRevise.com

Diagram A

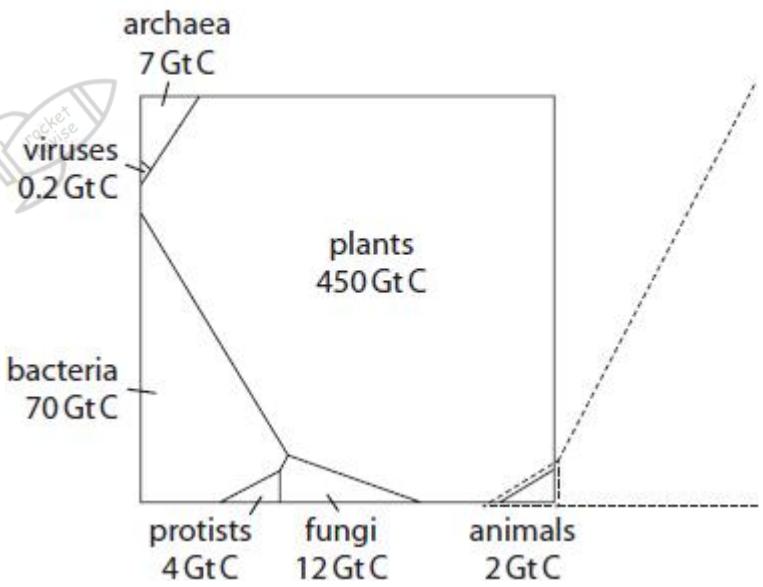
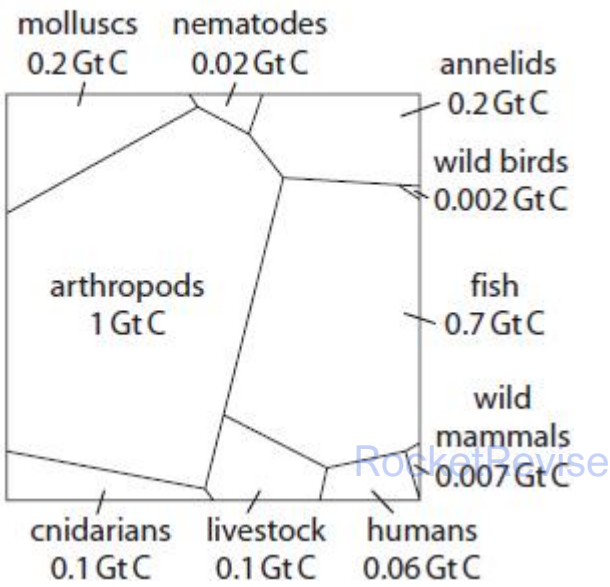


Diagram B



The area of each polygon is proportional to the biomass of that organism. The shape of each polygon has no meaning.

(i) Calculate the percentage of biomass in organisms belonging to the domain Eukarya.
Use the information in diagram A.

(2)

Answer ..... %

(ii) Suggest why the scientists studied the distribution of biomass in groups of organisms and not the number of individual organisms.
Use the information in diagram A.

(2)

.....

.....

.....

.....

.....

.....

(iii) Discuss the advantages and disadvantages of presenting data in Voronoi diagrams.
Use diagram B to support your answer.

(3)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

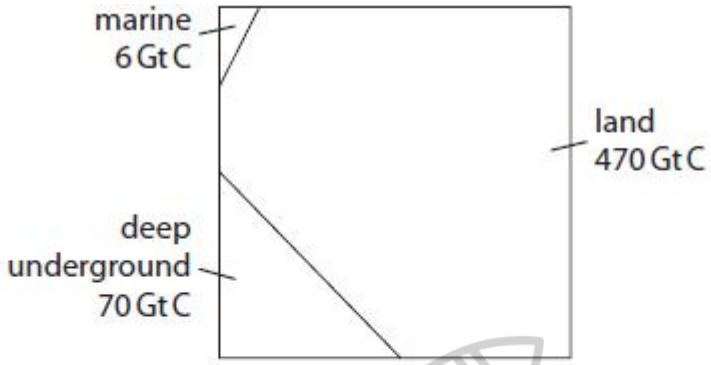
.....

.....

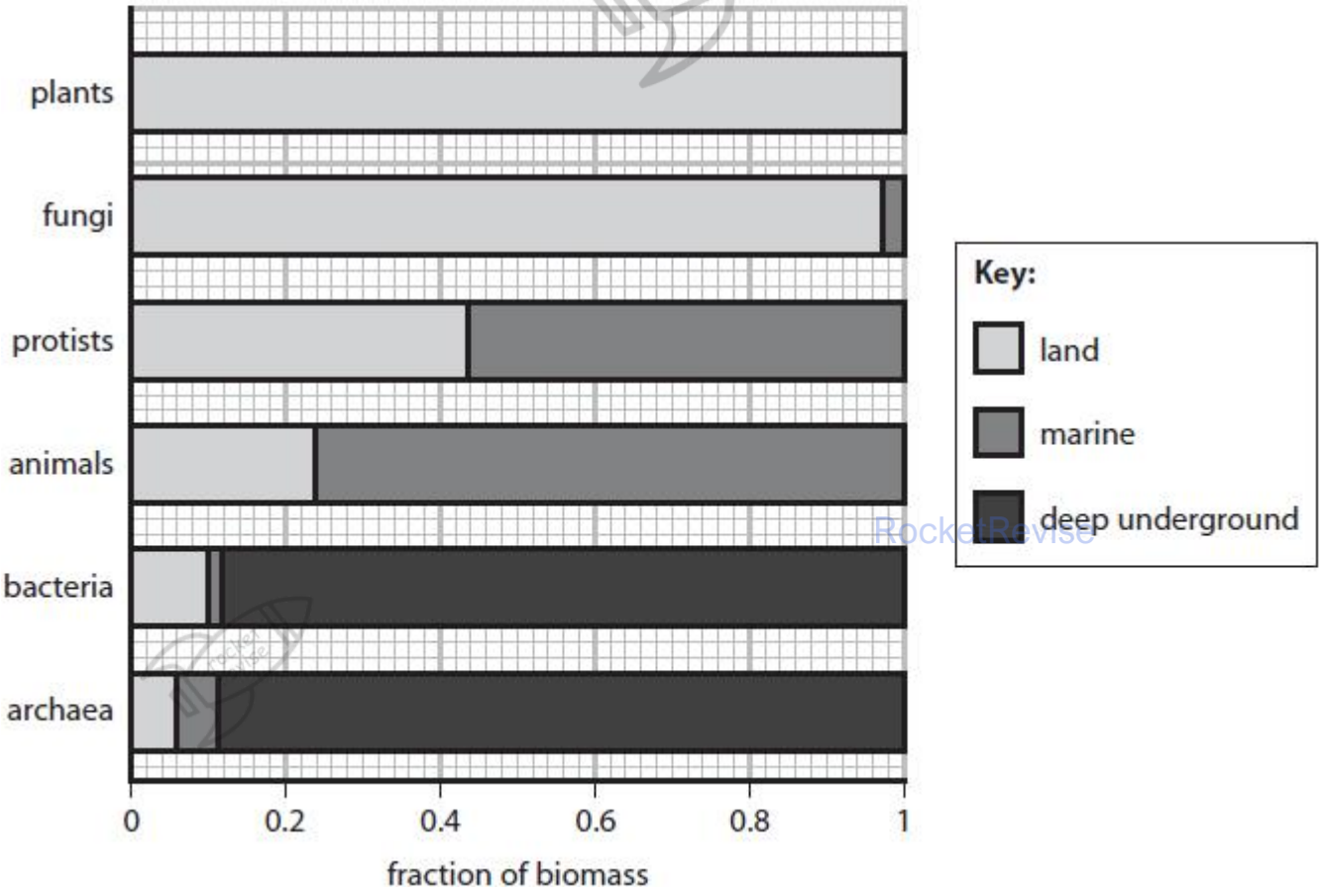
.....

* (c) The same study also determined the distribution of biomass in three different environments: marine, deep underground and land.

The Voronoi diagram shows the distribution of biomass in each environment.
The graph shows the proportion of biomass in different groups of organisms.



RocketRevise



RocketRevise

Explain the distribution of biomass in these three environments.
Use the information in the Voronoi diagram and the graph to support your answer.

(6)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 14 marks)

Q12.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Cultivation of seaweed for food is well-established in Asia.

(a) In 2017, the global harvest of seaweed was 32.9 million tonnes. Of this, 96.6% was cultivated.

Calculate the mass of cultivated seaweed.
Give your answer in standard form.

(1)

Answer ..... tonnes

www.RocketRevise.com

www.RocketRevise.com

(b) Seaweeds are photosynthetic organisms.

Photosynthesis can be described as a reaction requiring light to split apart the strong bonds in water molecules, storing the hydrogen in a fuel by combining it with carbon dioxide and releasing oxygen into the atmosphere.

(i) Which type of bond is broken in a water molecule?

(1)

- ☒ **A** covalent
- ☒ **B** glycosidic
- ☒ **C** hydrogen
- ☒ **D** ionic

(ii) Which is a fuel that stores hydrogen in plants and seaweeds?

(1)

- ☒ **A** ATP
- ☒ **B** glucose
- ☒ **C** glycogen
- ☒ **D** RUBISCO

RocketRevise

(iii) Which row of the table shows the reactions that combine hydrogen with carbon dioxide and release oxygen into the atmosphere?

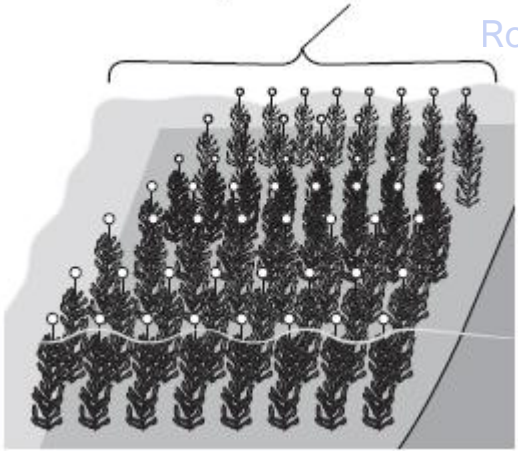
(1)

	Hydrogen is combined with carbon dioxide	Oxygen is released into the atmosphere
☒ **A**	light-dependent reactions	light-dependent reactions
☒ **B**	light-dependent reactions	light-independent reactions
☒ **C**	light-independent reactions	light-dependent reactions
☒ **D**	light-independent reactions	light-independent reactions

*(c) In Europe, cultivation of seaweed takes place on a small scale.

Large-scale cultivation of seaweed is now being considered.
Effective management of the conflict between human needs and conservation is necessary before large-scale cultivation can be used.
The diagram shows seaweed grown by suspended cultivation, a method used in Asia.

Rows of seaweed suspended on wires in the sea

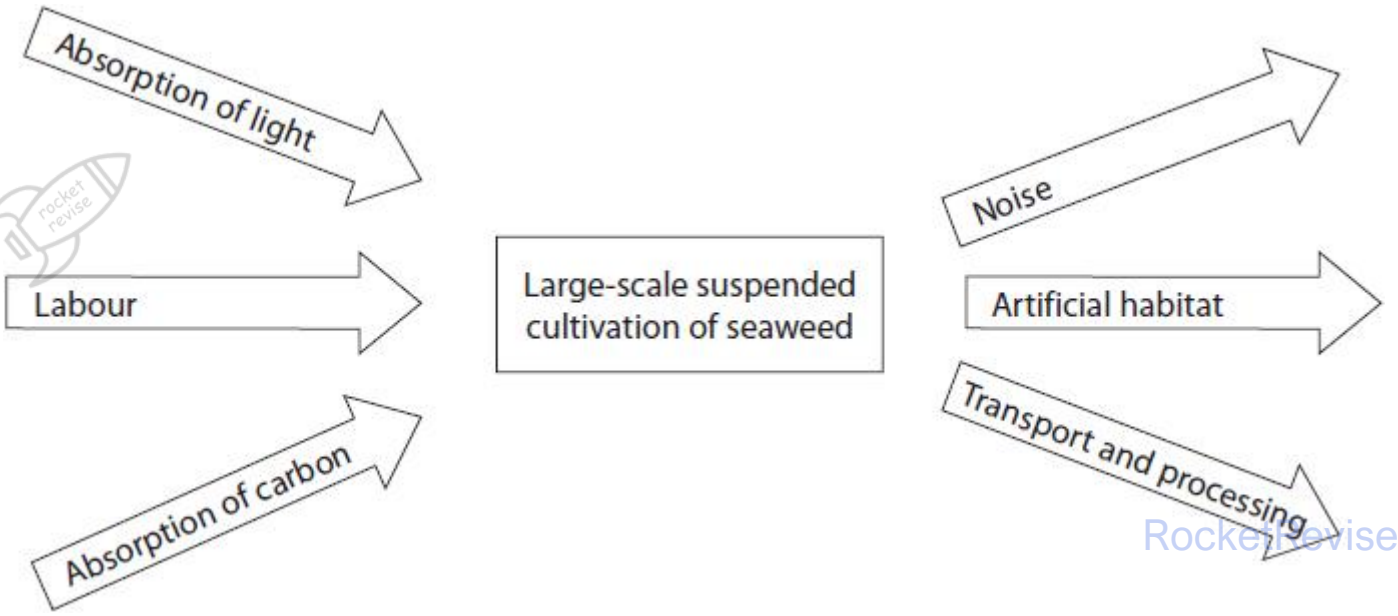


RocketRevise

The diagram shows some of the factors that have to be considered when managing the conflict between human needs and conservation.

www.RocketRevise.com

www.RocketRevise.com



Discuss the considerations that need to be taken into account when deciding whether or not large-scale seaweed production is suitable in Europe.

Use the information in the two diagrams and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 10 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Stages of succession can be seen on different parts of a sand dune system.

The greater the distance of the sand dune from the sea, the later the stage of succession.

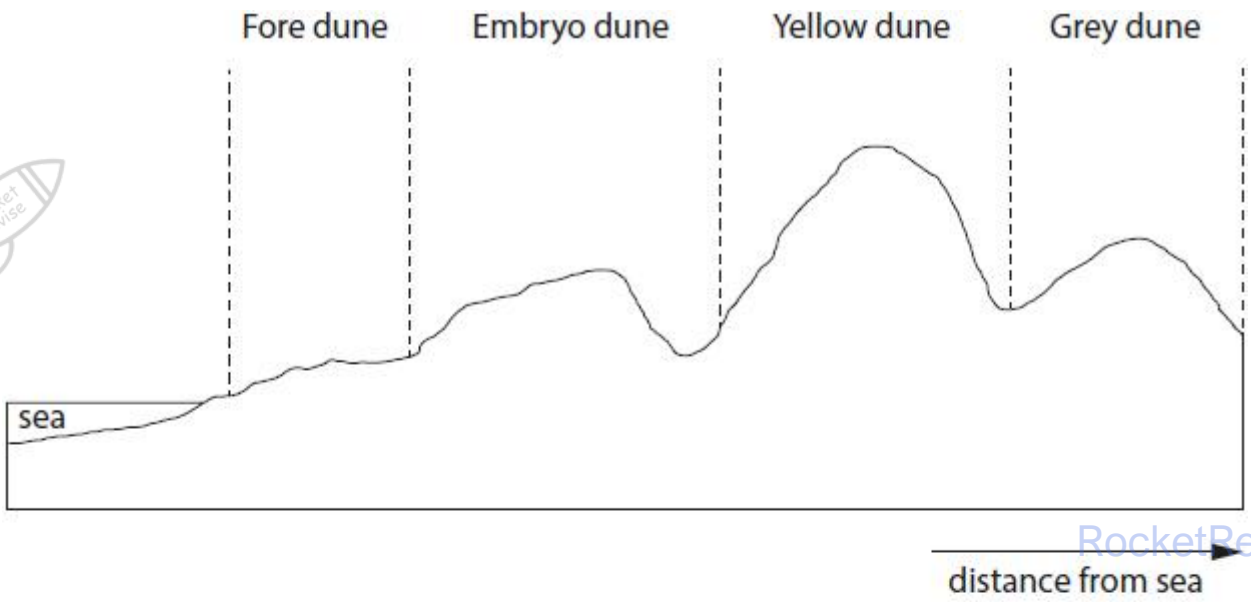
RocketRevise

The diagram shows a sand dune system and the table gives some information about the different types of sand dune in this system.

RocketRevise

www.RocketRevise.com

www.RocketRevise.com



Information	Fore dune	Embryo dune	Yellow dune	Grey dune
Soil depth / cm	<0.5	0.5	1.0	8.0
Percentage of organic matter (%)	<1.0	1.0	2.5	5.0
pH	8.5	8.0	7.0	6.5
Percentage of bare ground (%)	>97	97	70	10
Number of different plant species	2	3	6	15
Typical plant species	sea rocket saltwort	sea couch grass lyme grass marram grass	marram grass red fescue grass sea holly	range of different meadow plants

(a) (i) The pH of a solution is a measure of the concentration of hydrogen ions in that solution.

It is a log scale e.g. a solution of pH 5 contains 10^{-5} mol dm^{-3} of hydrogen ions.
What is the difference in concentration of hydrogen ions in a fore dune compared with a grey dune?

(1)

- ☒ **A** a fore dune has 2 times more hydrogen ions than the grey dune
- ☒ **B** a fore dune has 2 times fewer hydrogen ions than the grey dune
- ☒ **C** a fore dune has 100 times more hydrogen ions than the grey dune
- ☒ **D** a fore dune has 100 times fewer hydrogen ions than the grey dune

www.RocketRevise.com

www.RocketRevise.com

(ii) Calculate the percentage increase in the number of plant species on the grey dune compared with the fore dune.

(1)

Answer ..... %

*(iii) Explain the changes in these sand dunes with distance from the sea.
Use the information in the table to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

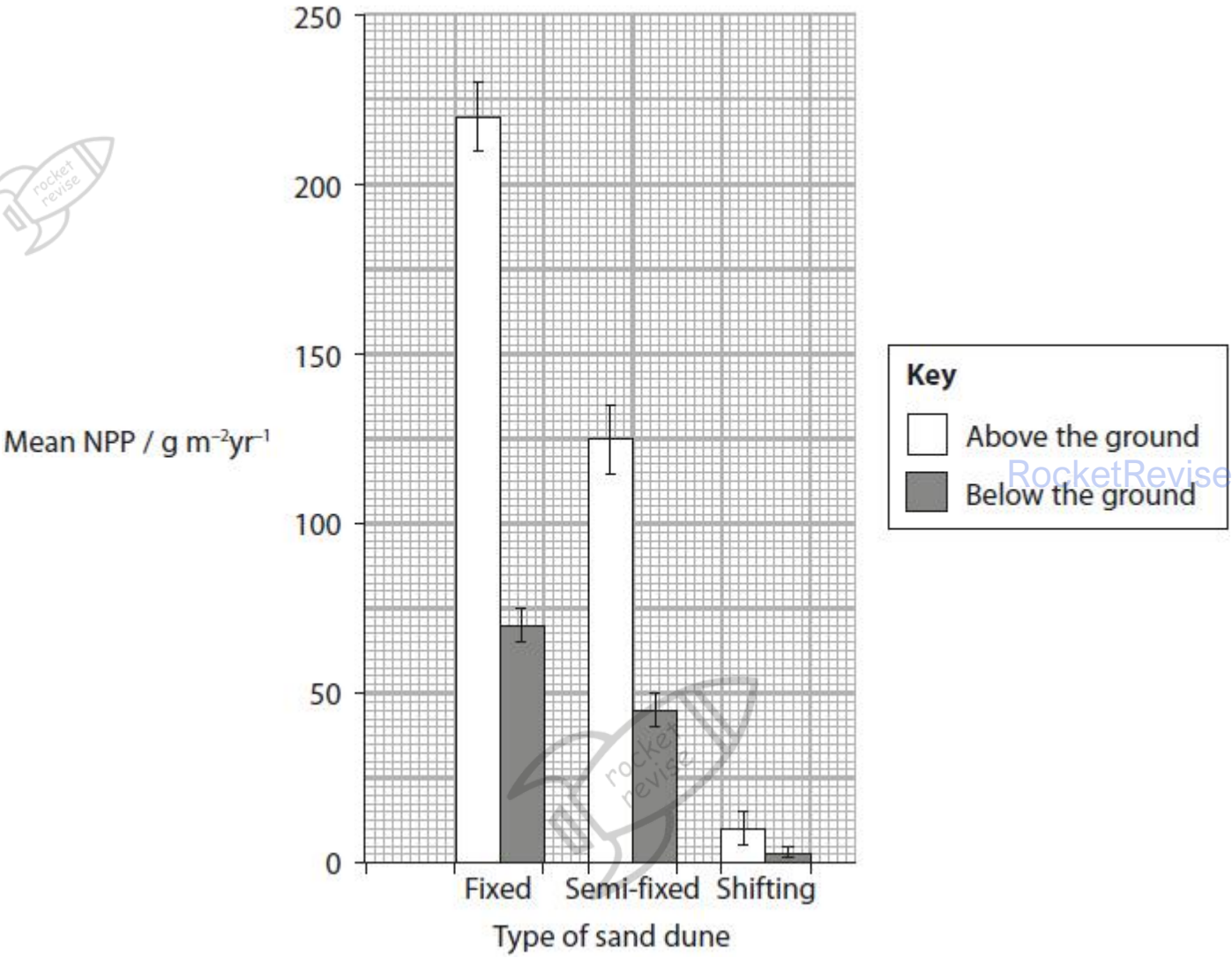
.....

(b) Sand dunes can be classified as fixed dunes, semi-fixed dunes and shifting dunes.

The graph shows the mean net primary productivity (NPP) above the ground and below the ground for these three types of sand dune.

www.RocketRevise.com

www.RocketRevise.com



(i) Describe the conclusions that can be drawn from this graph.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Explain how the products of the light-independent reactions become NPP below the ground.

(4)

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

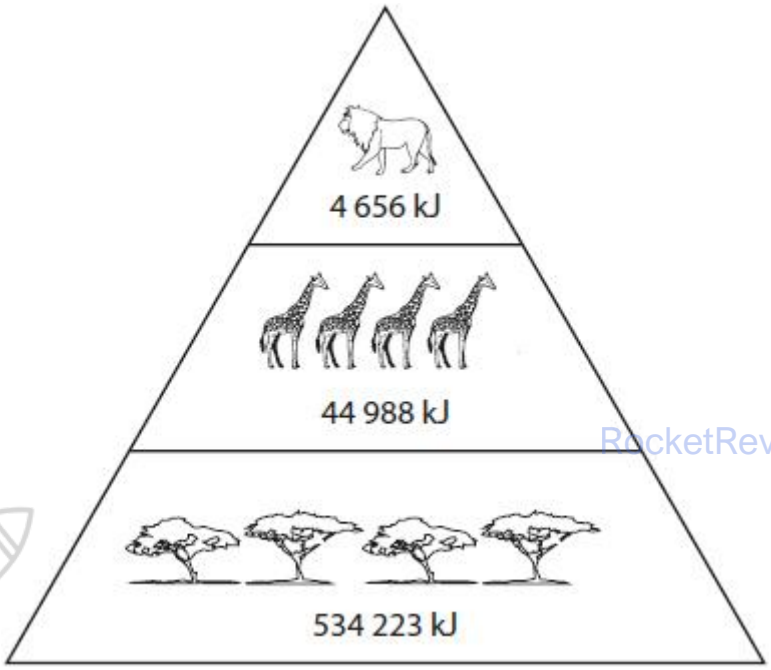
RocketRevise

(Total for question = 15 marks)

Q14.

Lions, giraffes and acacia trees are found on the African Plains.

The diagram shows the energy in the trophic levels that these organisms occupy.



RocketRevise

(a) Give the meaning of each of the following terms, using the information in the diagram to illustrate your answer.

(i) Habitat

(2)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

(ii) Population

(2)

.....

.....

.....

.....

.....

.....

(iii) Community

(2)

.....

.....

.....

.....

.....

.....

(iv) Niche

(2)

.....

.....

.....

.....

.....

.....

(b) Calculate the efficiency of energy transfer between trophic level one and trophic level two.

(1)

www.RocketRevise.com

www.RocketRevise.com

Answer ..... %

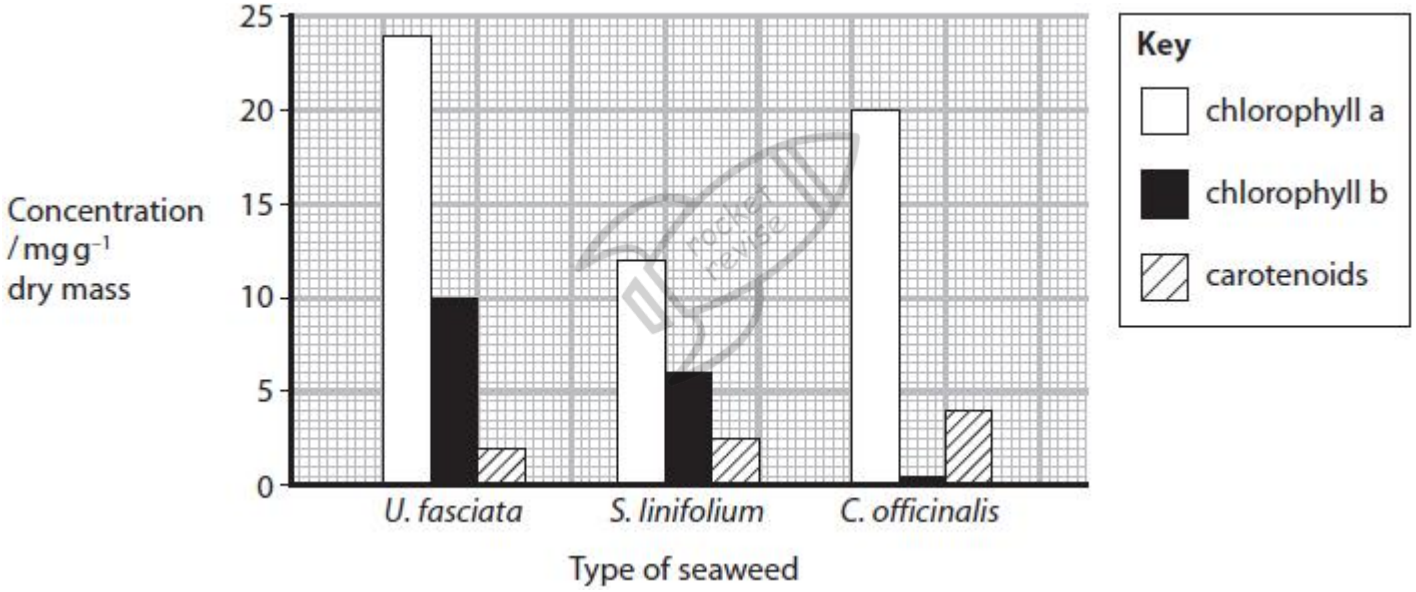
(Total for question = 9 marks)

Q15.

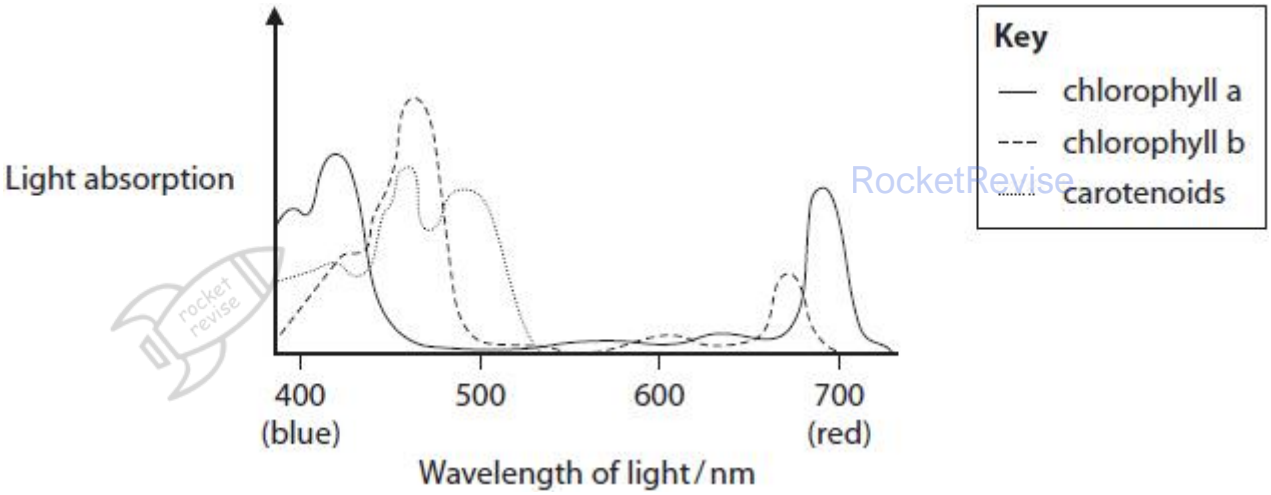
The composition of three seaweeds, *U. fasciata*, *S. linifolium* and *C. officinalis*, collected from a beach in Egypt, was determined.

RocketRevise

(a) The graph shows the chloroplast pigments extracted from these seaweeds.



The graph shows the absorption spectrum for these three chloroplast pigments.



(i) Which row of the table shows the colour of each of these seaweeds?

(1)

www.RocketRevise.com

www.RocketRevise.com

	U. fasciata	S. linifolium	C. officinalis
☐ A	green	brown	red
☐ B	green	red	green
☐ C	red	brown	red
☐ D	red	green	green

(ii) Explain the role of chloroplast pigments in the light-dependent reactions of photosynthesis.

(2)

.....

.....

.....

.....

.....

.....

(b) The table shows some of the compounds found in two of these seaweeds.

Compound		U. fasciata	S. linifolium
*carotenoids	/ mg g^{-1} dry mass	2.00	3.00
*phenols	/ a.u. g^{-1} dry mass	11.95	10.35
*flavonoids	/ a.u. g^{-1} dry mass	7.04	4.53
carbohydrates	/ percentage of dry mass	23.70	25.03
proteins	/ percentage of dry mass	9.56	14.89
lipids	/ percentage of dry mass	2.96	2.16

* antioxidants

(i) Describe how GALP, produced in chloroplasts, is converted into protein.

(3)

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

*(ii) These seaweeds are considered to be a healthy, environmentally-friendly and economic food source.

Comment on this statement.

Use the information in the table and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 12 marks)

Q16.

The photograph shows a ground squirrel.

www.RocketRevise.com

www.RocketRevise.com



(Source: Caroline Wilcox)

magnification × 0.5

Ground squirrels feed on a variety of foods including nuts.

Ground squirrels have evolved food pouches.

The photograph shows the food pouches of a ground squirrel.



(Source: © All Canada Photos/Alamy Stock Photo)

The table shows some information about three types of nut: acorns, hazelnuts and walnuts.

www.RocketRevise.com

www.RocketRevise.com

Type of nut	Photograph	Description	Energy content / kJ per 100 g
acorn		1.0 to 3.0 cm long 0.8 to 2.0 cm wide	1619
hazelnut		1.5 to 2.5 cm long 1.0 to 1.5 cm wide hard covering	2788
walnut		3.0 to 5.0 cm long 2.0 to 4.0 cm wide hard shell	2822

(Source: © Science photo library, Source: © Jon Stokes/Science photo library, Source: © Science photo library)

(a) The volume of the smallest walnut is approximately 50 cm^3.

Calculate the approximate volume of the largest hazelnut, using the formula

$$\text{volume} = \frac{4}{3}\pi l w^2$$

where $\pi = 3$, l = length of the hazelnut and w = width of the hazelnut.

(1)

RocketRevise

Answer ..... cm^3

(b) A student investigated the feeding preferences of ground squirrels.

(i) Design a laboratory investigation to find out which of these three types of nut ground squirrels prefer to eat.

(3)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Predict the results of this investigation, giving reasons.
Use the information in the table to support your answer.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) Suggest how ground squirrels evolved pouches.

(3)

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

(Total for question = 10 marks)

Q17.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

Redwood trees are the tallest living organisms on Earth. Some of the older trees are more than 2 000 years old.

Dendrochronology can be used to work out how old a tree is and how much it has grown each year.

A sample of the tree trunk core can be taken using an increment borer.

The photograph shows an increment borer being used to take a sample of a tree trunk core.

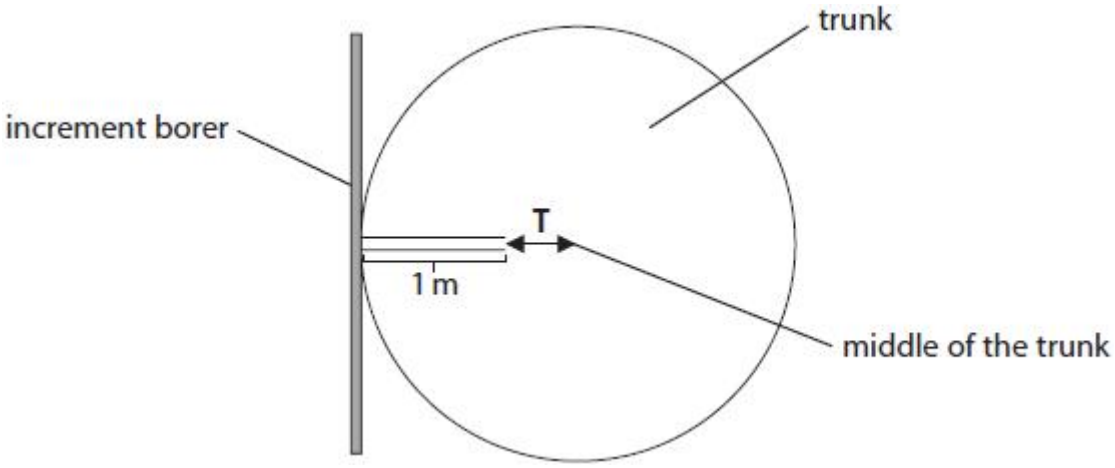


(Source: © Custom Life Science Images / Alamy Stock Photo)

(a) The diagram shows an increment borer pushed into the trunk of a large Redwood tree.

www.RocketRevise.com

www.RocketRevise.com



An increment borer 1 m long was inserted into a Redwood tree. The borer did not reach the middle of the trunk. The distance from the end of the borer to the middle of the trunk is **T** cm.

(i) Calculate the radius (r) of a tree with a circumference (C) of 8 m.

Use the formula: $r = \frac{C}{2\pi}$

Give your answer to **two** decimal places.

(1)

Answer ..... m

(ii) Calculate the distance **T** as a percentage of this radius.

(1)

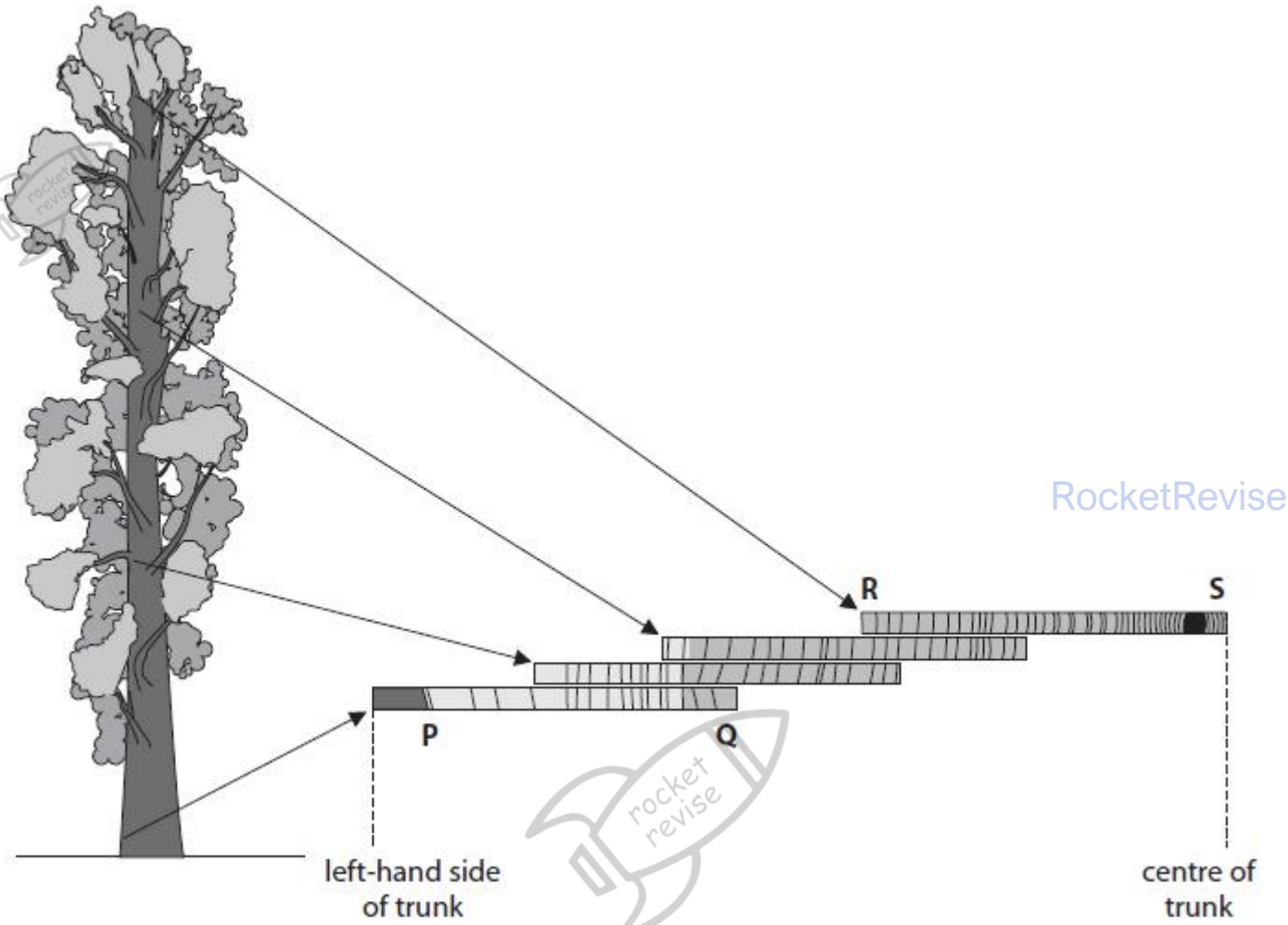
Answer ..... %

(b) To calculate how old a tree is and how much it has grown, core samples are taken at different heights up the tree.

These core samples are then aligned by matching the rings, as shown in the diagram.

www.RocketRevise.com

www.RocketRevise.com



(i) Which row of the table shows the newest and oldest rings in this tree?

(1)

		newest ring	oldest ring
☒	A	P	Q
☒	B	P	S
☒	C	S	R
☒	D	S	P

(ii) Explain how these core samples can be used to calculate the age of this tree.
Use the information in the question and the diagram to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

(iii) Describe how the growth rate of this tree can be calculated.

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 8 marks)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Plants store biomass both above and below ground.

(a) Scientists have found that on average, 24 % of plant biomass is underground in the roots.

The total plant biomass underground contains 113 gigatonnes of carbon. This is equivalent to the mass of carbon dioxide emissions produced by humans in 10 years.
One gigatonne is 1×10^{12} kg.
Calculate the mass of carbon dioxide, in kilograms, produced by humans in one year.
Give your answer in standard form.

(2)

RocketRevise

Answer ..... kg

(b) Describe how GALP, produced in the leaves, becomes incorporated into biomass in the roots.

(4)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(c) Inorganic ions are used by plants to make molecules.

The table shows some molecules made by plants.
For each molecule, put one cross ☒ in the appropriate box, in each row, to show which inorganic ion provides an atom found in the molecule.

(3)

Molecule	Inorganic ion			
	nitrate only	phosphate only	both nitrate and phosphate	neither nitrate nor phosphate
cellulose	☐	☐	☐	☐
nucleic acid	☐	☐	☐	☐
triglyceride	☐	☐	☐	☐

* (d) Forests, shrublands and grasslands are three types of ecosystem.

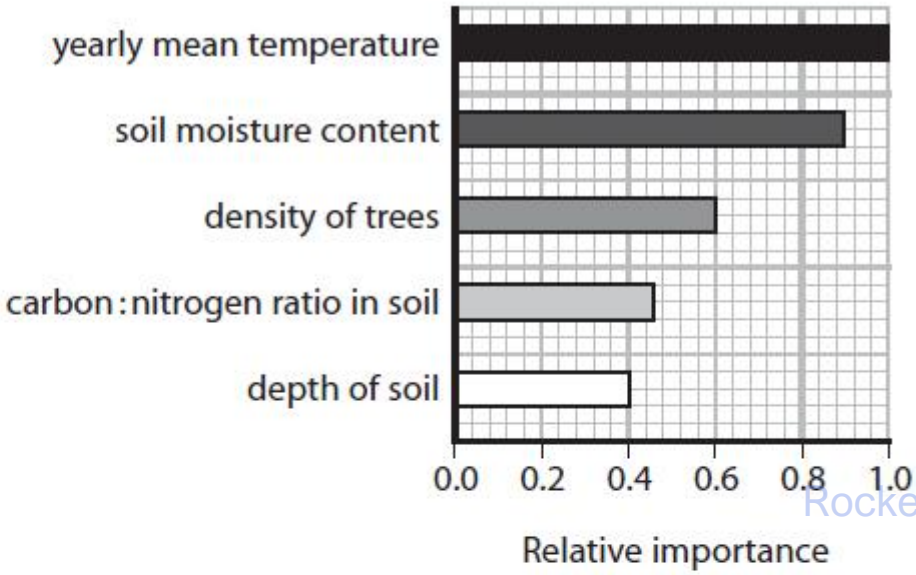
Biotic and abiotic factors affect how much biomass is stored in the roots of plants.
The graphs show the importance of some abiotic factors in determining how much biomass is stored in the roots in these ecosystems.

RocketRevise

www.RocketRevise.com

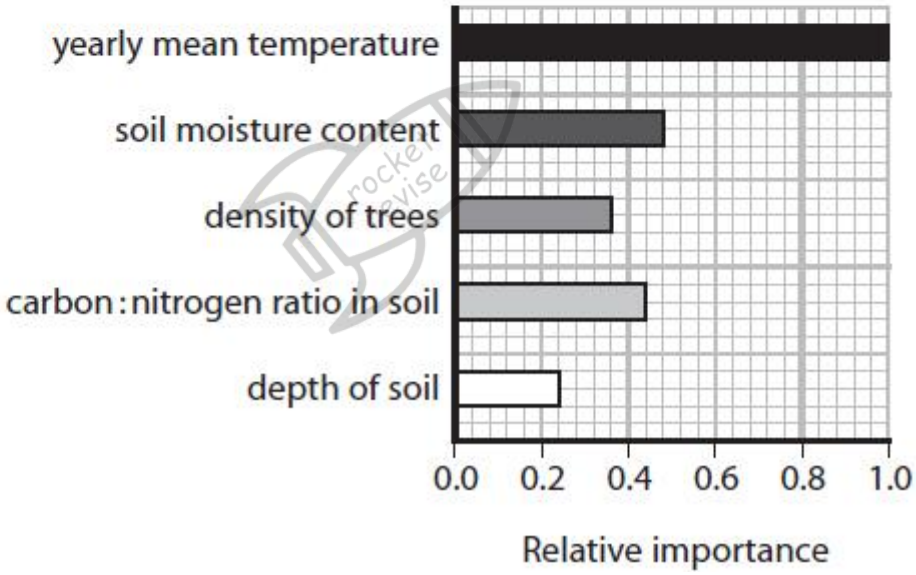
www.RocketRevise.com

Forests

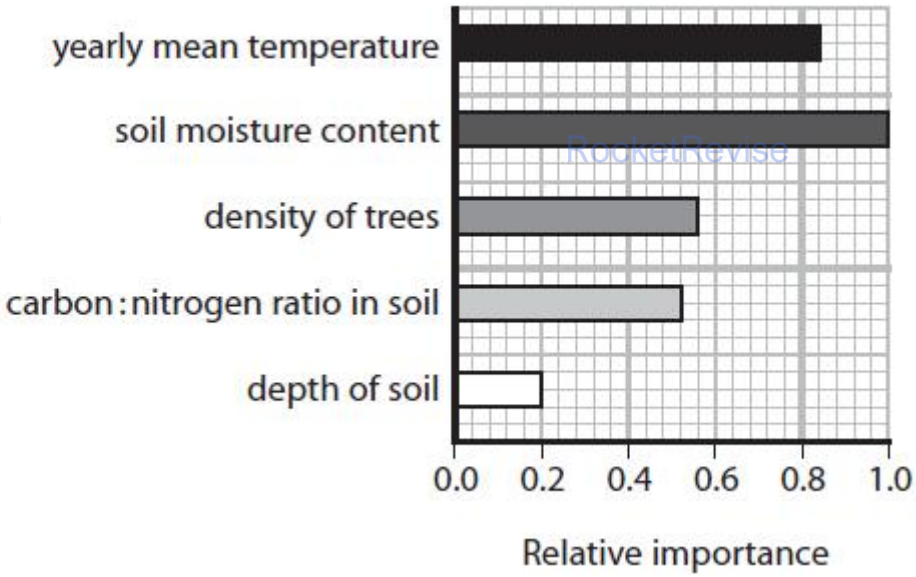


RocketRevise

Shrublands



Grasslands



RocketRevise

Discuss how these factors could affect the proportion of biomass stored in the roots of plants growing in these three ecosystems.

(6)

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 15 marks)

Q19.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

(a) The table gives some information about structures found in chloroplasts.

www.RocketRevise.com

www.RocketRevise.com

Structure	Function	Other information
P	protein synthesis	some are 20 nm in size
Q	site of light-independent reactions	colourless
R	compartment for accumulation of hydrogen ions	largest are 4.35×10^{-4} mm long
S	storage granule	1 to 35 μm in size

(i) Where in the chloroplast is RUBISCO active?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(ii) What is stored in S?

(1)

- ☐ **A** GALP
- ☐ **B** lipid
- ☐ **C** starch
- ☐ **D** sucrose

(iii) Where is GP formed?

(1)

- ☐ **A** P
- ☐ **B** Q
- ☐ **C** R
- ☐ **D** S

(iv) Which is the correct order of size, smallest to largest?

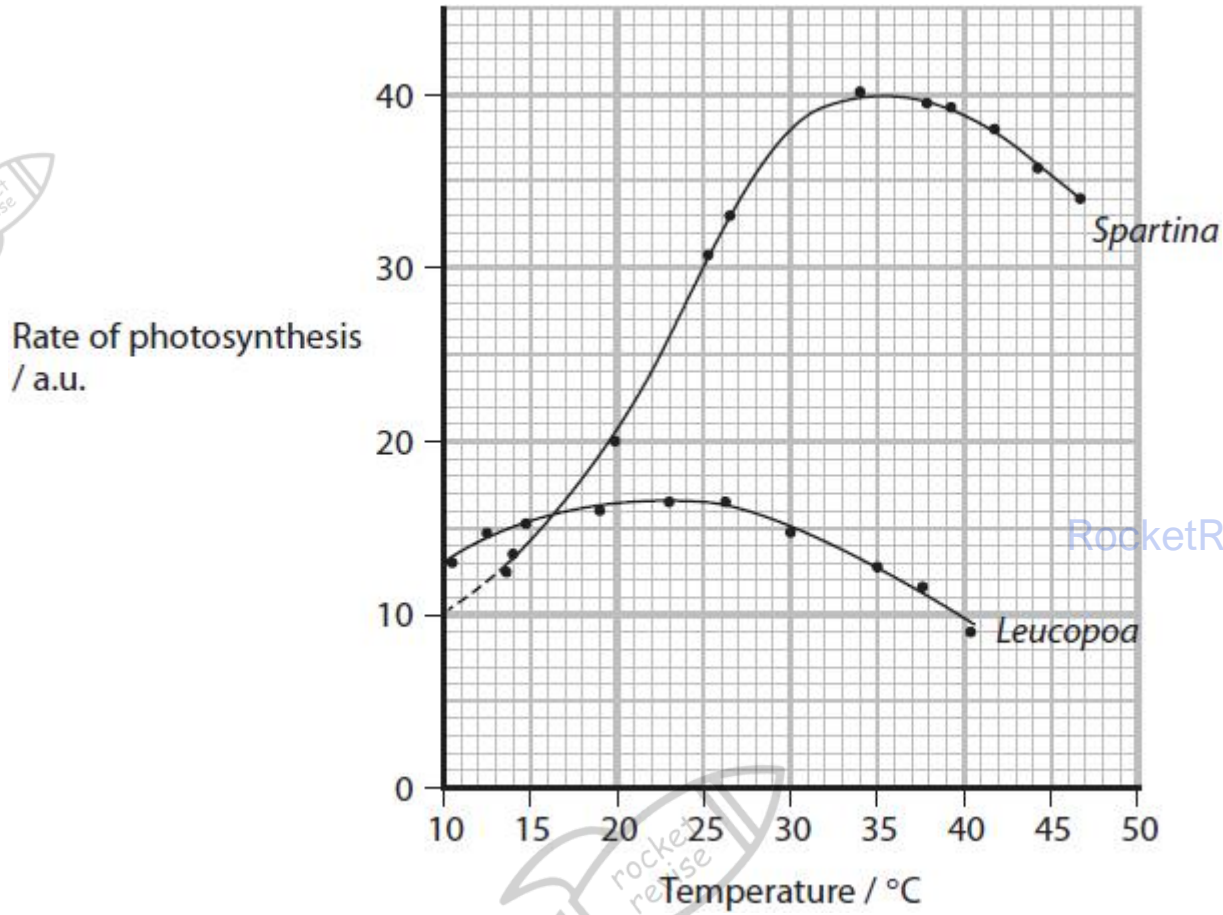
(1)

- ☐ **A** P R S
- ☐ **B** P S R
- ☐ **C** R S P
- ☐ **D** S R P

(b) The graph shows the rate of photosynthesis in two types of plant, *Spartina* and *Leucopoa*, at different temperatures.

www.RocketRevise.com

www.RocketRevise.com



(i) Which could be the units for rate of photosynthesis?

(1)

- ☒ **A** mg CO_2 produced mm^{-2} hr^{-1}
- ☒ **B** mg CO_2 produced mm^{-1} hr^{-2}
- ☒ **C** mg CO_2 used mm^{-2} hr^{-1}
- ☒ **D** mg CO_2 used mm^{-1} hr^{-2}

(ii) Compare and contrast the rate of photosynthesis of these two species of plant, at different temperatures.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(iii) Describe how a Q_{10} value for the rate of photosynthesis can be calculated, using this graph.

(2)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

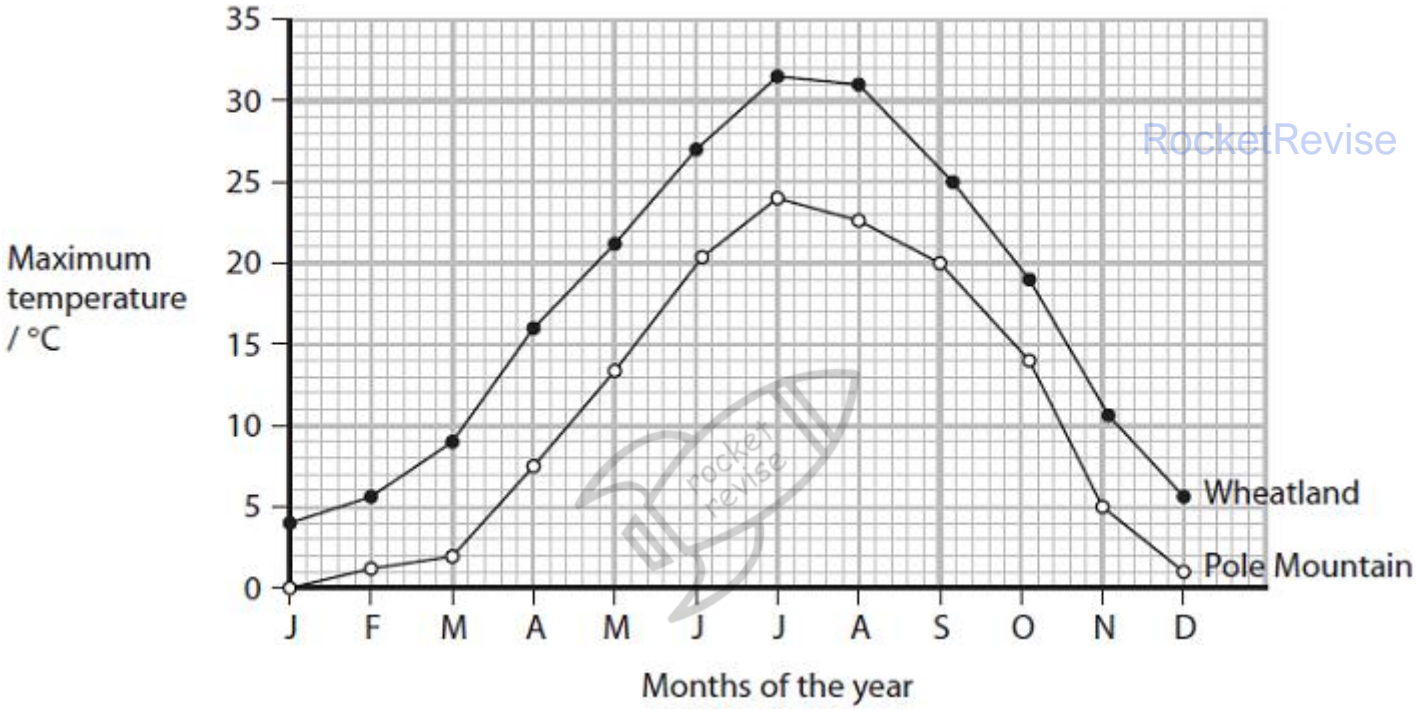
.....

.....

.....

.....

(iv) The graph shows the maximum temperature, for each month, in two regions of North America: Wheatland and Pole Mountain.



Explain in which of these two regions of North America *Spartina* is most likely to grow. Use the information in both graphs to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 13 marks)

www.RocketRevise.com

www.RocketRevise.com

Q20.

There has been an increase in the number of forest fires in many areas of the world.

Climate change caused by humans is claimed to be responsible for this.

The photograph shows the devastation caused by a forest fire in Myanmar.



(Source: © robertharding/Alamy Stock Photo)

(a) (i) State the term used to describe climate change caused by human activity.

(1)

.....

(ii) Explain why some people consider this claim to be controversial.

(2)

.....

.....

.....

.....

.....

.....

* (b) Discuss the effects that forest fires have on the local atmosphere and local biodiversity, in the short term and the long term.

(6)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) Reforestation is one way to repair the damage caused to forests by climate change.

Explain why the changes to the environment caused by climate change need to be considered when selecting the species of trees to use in reforestation.

(4)

.....

.....

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

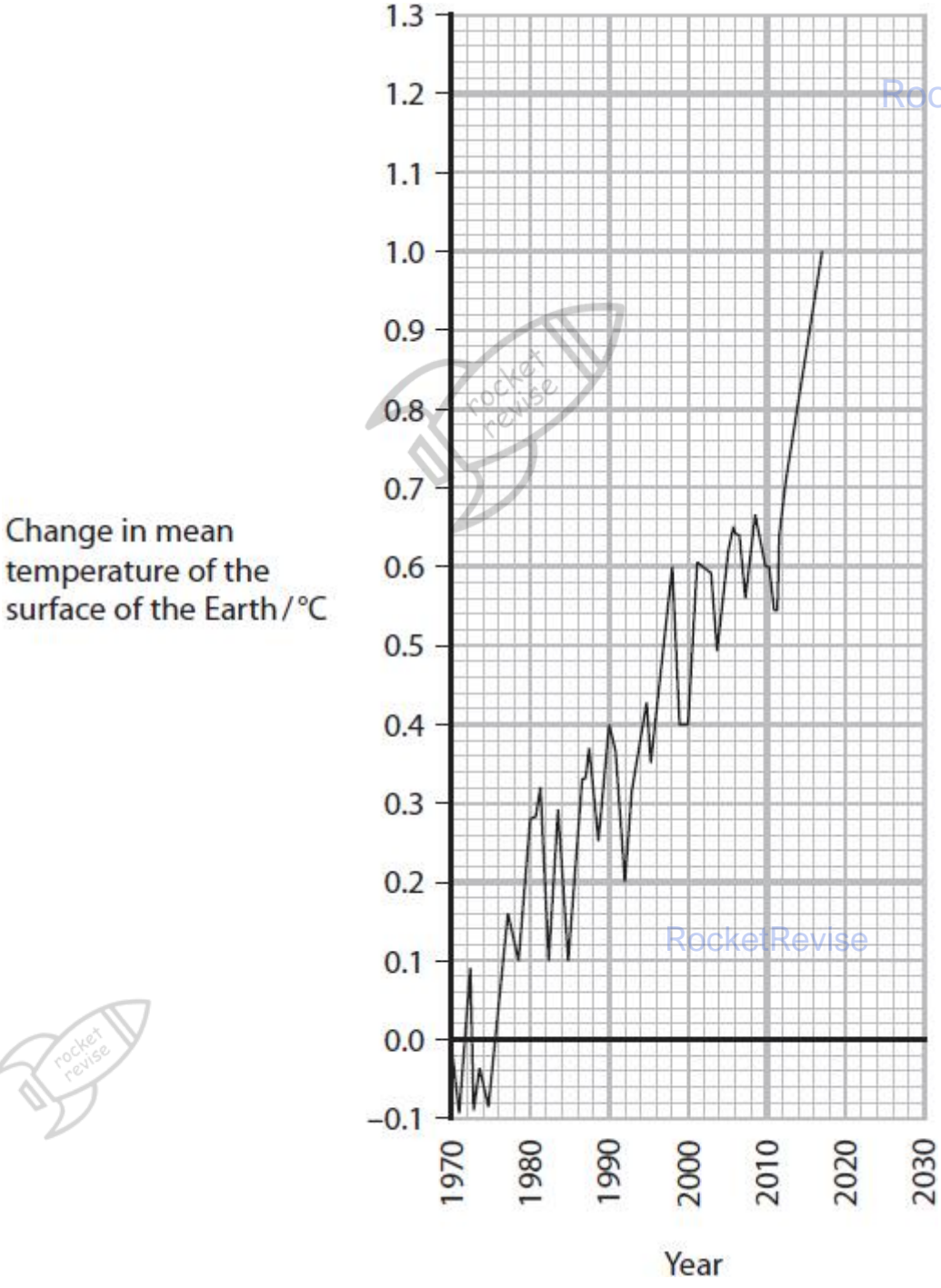
www.RocketRevise.com

.....

(Total for question = 13 marks)

Q21.

The graph shows the changes in mean temperature of the surface of the Earth from 1970 to 2017.



(a) Which of the following gases could be responsible for these temperature changes?

(1)

☒ **A** carbon dioxide and methane

www.RocketRevise.com

www.RocketRevise.com

☒ **B** carbon dioxide and oxygen

☒ **C** carbon dioxide, methane and oxygen

☒ **D** methane and oxygen

(b) What is the estimated change in the mean temperature of the surface of the Earth in 2030?

(1)

☒ **A** 1.3

☒ **B** 1.1

☒ **C** 0.8

RocketRevise

☒ **D** 0.2

(c) Ultraviolet light has a shorter wavelength than infrared radiation.

Which row of the table describes the passage of ultraviolet light and infrared radiation, reflected from the surface of the Earth, through gases in the atmosphere?

(1)

	Ultraviolet light	Infrared radiation
☒ **A**	passes through	passes through
☒ **B**	passes through	does not pass through
☒ **C**	does not pass through	passes through
☒ **D**	does not pass through	does not pass through

(d) Studying the width of tree rings provides evidence for climate change.

(i) Which of the following involves the study of tree rings?

(1)

☒ **A** dendrochronology

☒ **B** entomology

☒ **C** epigenetics

☒ **D** proteomics

RocketRevise

(ii) Explain why the width of tree rings produced in 1970 would be different from the width of tree rings produced in 2010.

(3)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

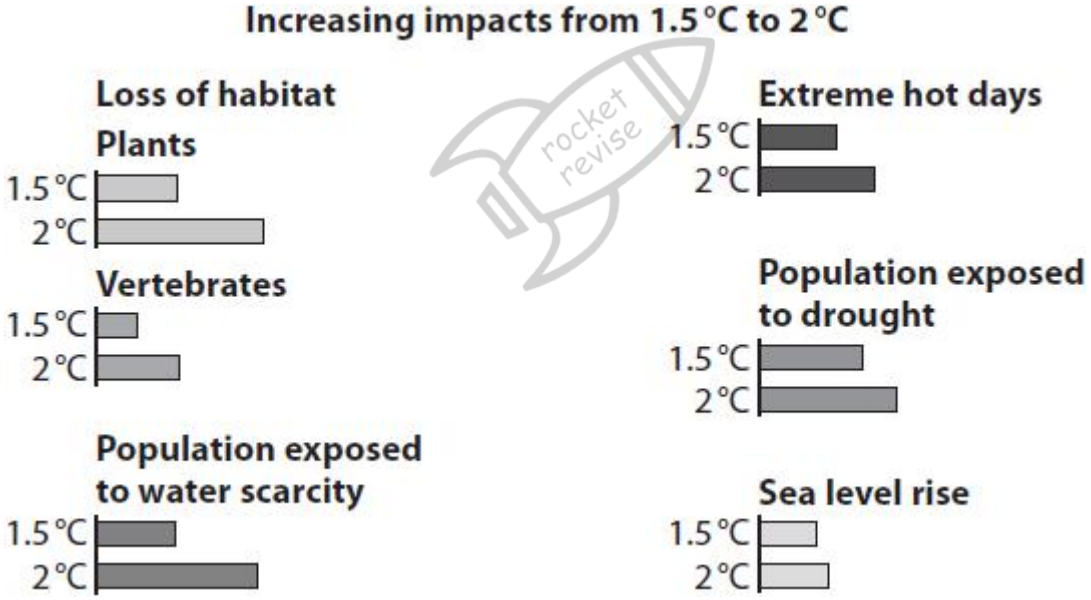
(Total for question = 7 marks)

Q22.

RocketRevise

Global warming is one consequence of anthropogenic climate change.

The diagram shows a model of some of the effects an increase in temperature of either 1.5 °C or 2 °C could have on the environment.



(a) (i) Explain why this diagram is described as a model of the effects of an increase in temperature.

(2)

RocketRevise

.....

.....

.....

.....

.....

.....

(ii) Calculate the percentage increase in the population exposed to drought if the temperature increased by 2 °C and not 1.5 °C.

Assume the bars in the diagram are drawn to scale.

(1)

www.RocketRevise.com

www.RocketRevise.com

Answer ..... %

* (b) It has been estimated that because humans have planted more trees and crops, the leaf area increased by 0.5 million km2 between 2019 and 2000.

China accounts for 25% of this increase. It planted forests and crops, in equal proportions.
India accounts for 7% of this increase. It planted mostly crops.
Discuss the possible impacts of planting forests or crops on global warming, biodiversity and the local population.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

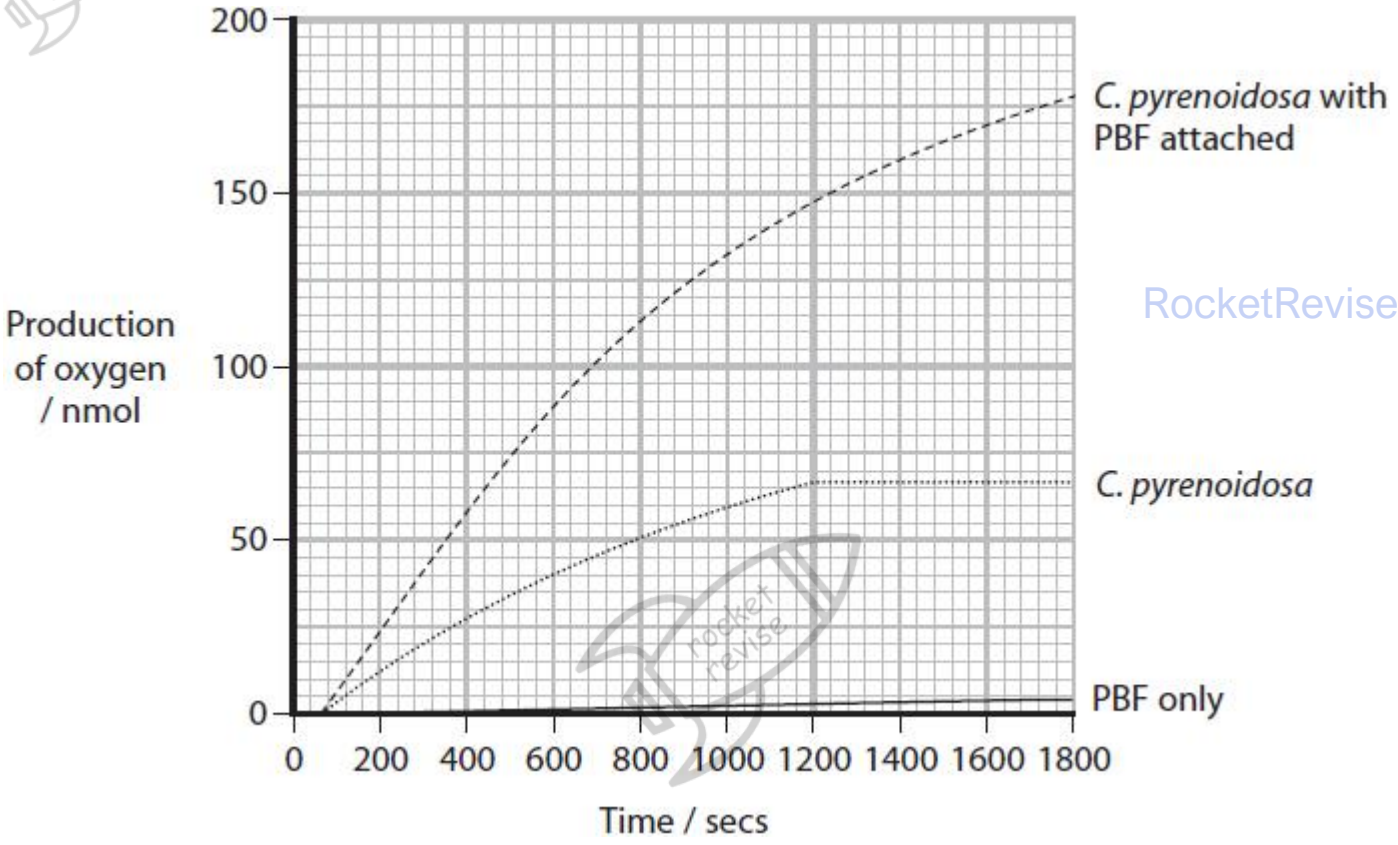
Q23.

www.RocketRevise.com

www.RocketRevise.com

(a) In an investigation, scientists attached a molecule called PBF to the cell surface of a species of green algae, *C. pyrenoidosa*.

These algae are photosynthetic organisms and PBF absorbs green light.
The scientists measured the production of oxygen by *C. pyrenoidosa*, *C. pyrenoidosa* with PBF attached and PBF only.
The graph shows the results of this investigation.



(i) Calculate the rate of production of oxygen at 1 200 seconds for *C. pyrenoidosa* with PBF attached.

(3)

Answer ..... nmol $secs^{-1}$

(ii) Explain the effect PBF has on the production of oxygen.

(3)

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(b) These scientists also measured the mass of *C. pyrenoidosa*, with and without PBF attached, after a 20-day period.

They found that the increase in mass of *C. pyrenoidosa* with PBF attached was greater than the increase in mass of *C. pyrenoidosa* without PBF attached. Explain these findings.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

Q24.

RocketRevise

Natural rubber is harvested from rubber trees.

The photograph shows rubber being collected from the trunk of a rubber tree in Thailand.

www.RocketRevise.com

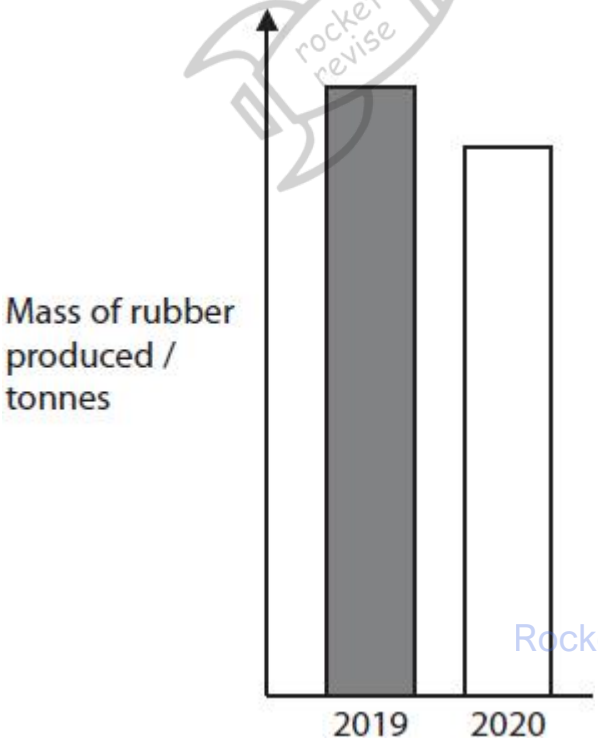
www.RocketRevise.com



RocketRevise

(Source: © Tong Patong/Alamy Stock Photo)

(a) The graph shows the mass of rubber produced in Thailand in 2019 and 2020.



RocketRevise

In 2020, Thailand produced 4.37 million tonnes of rubber.
Estimate the mass of rubber produced in Thailand in 2019.

(1)

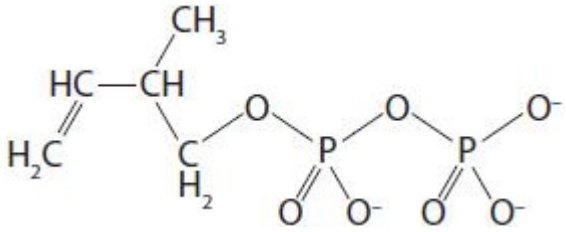
Answer ..... tonnes

(b) Rubber is synthesised by specialised cells in the trunks of rubber trees.

The diagram shows the structure of one molecule that these cells use to synthesise rubber.

www.RocketRevise.com

www.RocketRevise.com



GALP is produced in the light-independent reactions in the leaves.
Explain how the cells in the trunk are able to synthesise the molecule in the diagram using GALP produced in the leaves.
Use the information in the diagram and your own knowledge to support your answer.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(c) A rubber tree can be used to harvest rubber until it is about 32 years old.

Explain how dendrochronology could be used to determine the age of a tree that is still producing rubber.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

(Total for question = 8 marks)

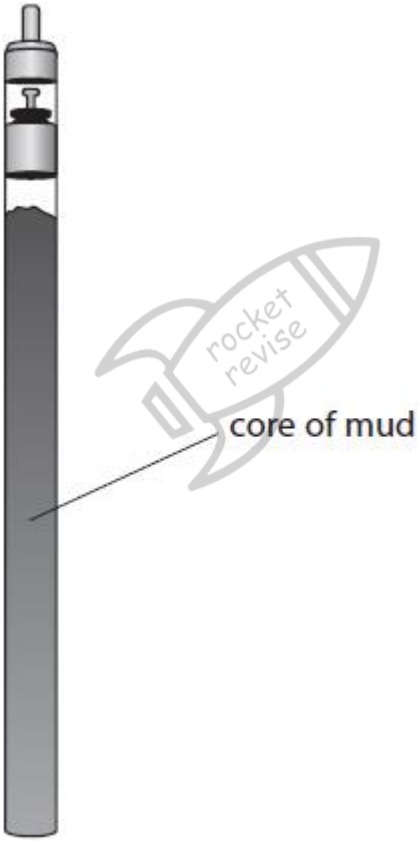
Q25.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Scientists have used specialised apparatus to remove cores of mud from a tropical rainforest in the Republic of the Congo.

The diagram shows the apparatus containing a core of mud.

RocketRevise



(a) The core of mud removed has a diameter of 80 mm and a length of 900 mm.

Which is the volume of this core of mud?
Use the formula: $V = \pi r^2 l$
Use $\pi = 3.142$

RocketRevise

(1)

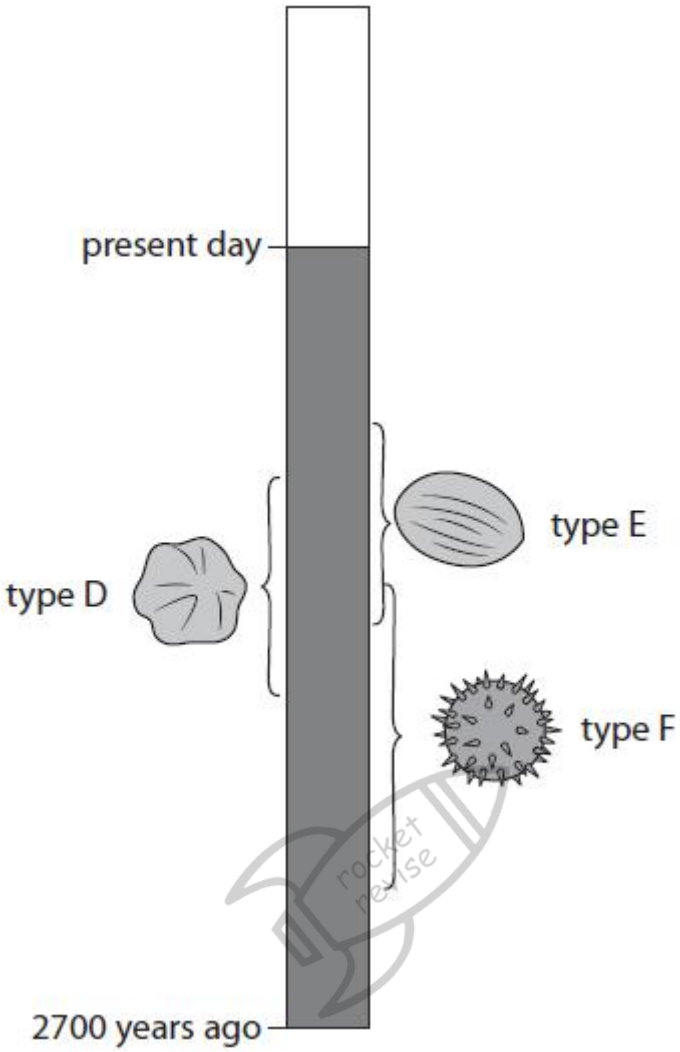
- ☒ **A** 4524 cm^3
- ☒ **B** 4525 cm^3
- ☒ **C** 18 097 cm^3
- ☒ **D** 18 098 cm^3

(b) The scale diagram shows where three types of pollen grain were found in a core of mud.

The depth of the mud is proportional to how long ago the mud was deposited.

www.RocketRevise.com

www.RocketRevise.com



RocketRevise

(i) Calculate how many years the plants producing type D pollen grains were present in this rainforest.

(2)

RocketRevise

Answer ..... years

(ii) Explain the distribution of these three types of pollen grain in this mud column.
Use the information in this diagram to support your answer.

(4)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

(Total for question = 7 marks)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

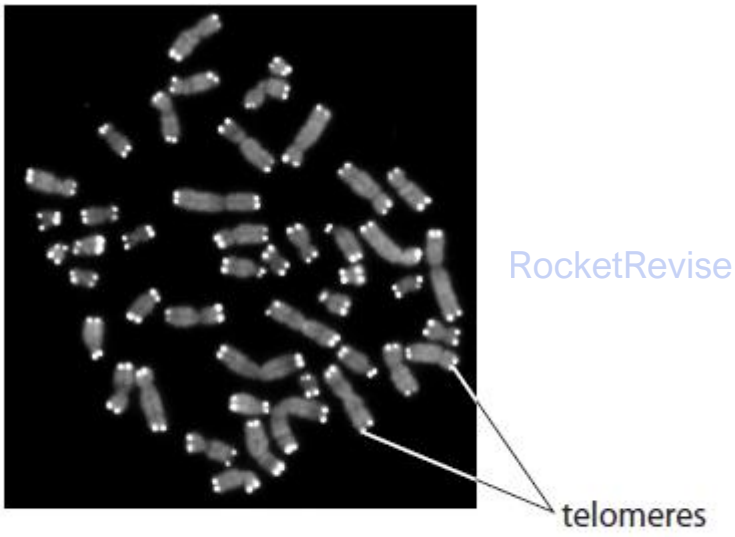
A telomere is a repetitive nucleotide sequence at the end of a chromosome.

The telomere protects the end of each chromosome from damage.

Each time DNA is synthesised the telomeres become shorter.

When the telomeres become too short the cell dies.

The photograph shows telomeres on the ends of chromosomes.



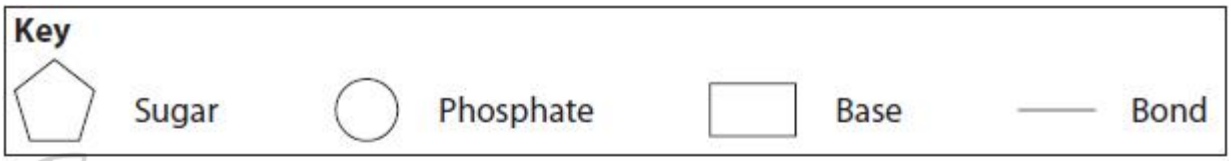
(Source: © Hesed Padilla-Nash and Thomas Ried, National Cancer Institute, National Institutes Of Health / Science Photo Library)

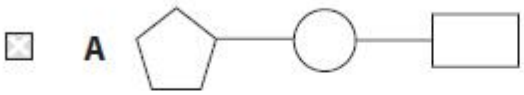
(a) (i) Which diagram shows a mononucleotide of DNA?

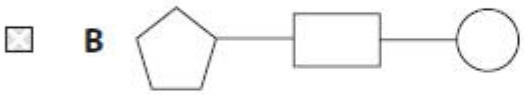
(1)

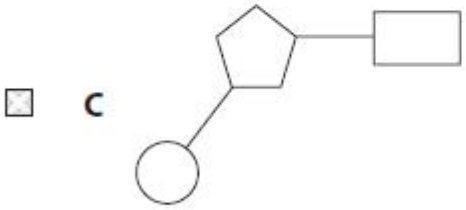
www.RocketRevise.com

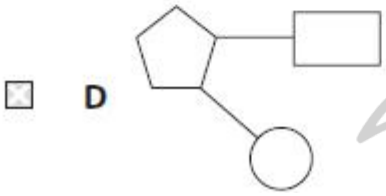
www.RocketRevise.com











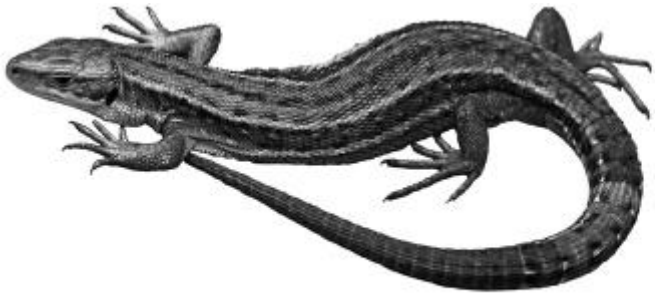
RocketRevise

(ii) Name the bond that joins nucleotides together in a strand of DNA.

(1)

.....

(b) The photograph shows a common lizard.



RocketRevise

(Source: © 'Nature cutout's / Alamy Stock Photo)

Lizards cannot regulate their body temperature and are at risk of extinction due to global warming.

The effect of temperature on the length of young lizards was investigated.

The change in abundance of lizards of all ages, over a 12 year period, was also investigated.

(i) There was a positive correlation between temperature and length of young lizards. This could be due to enzyme activity.

Explain this correlation.

(3)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

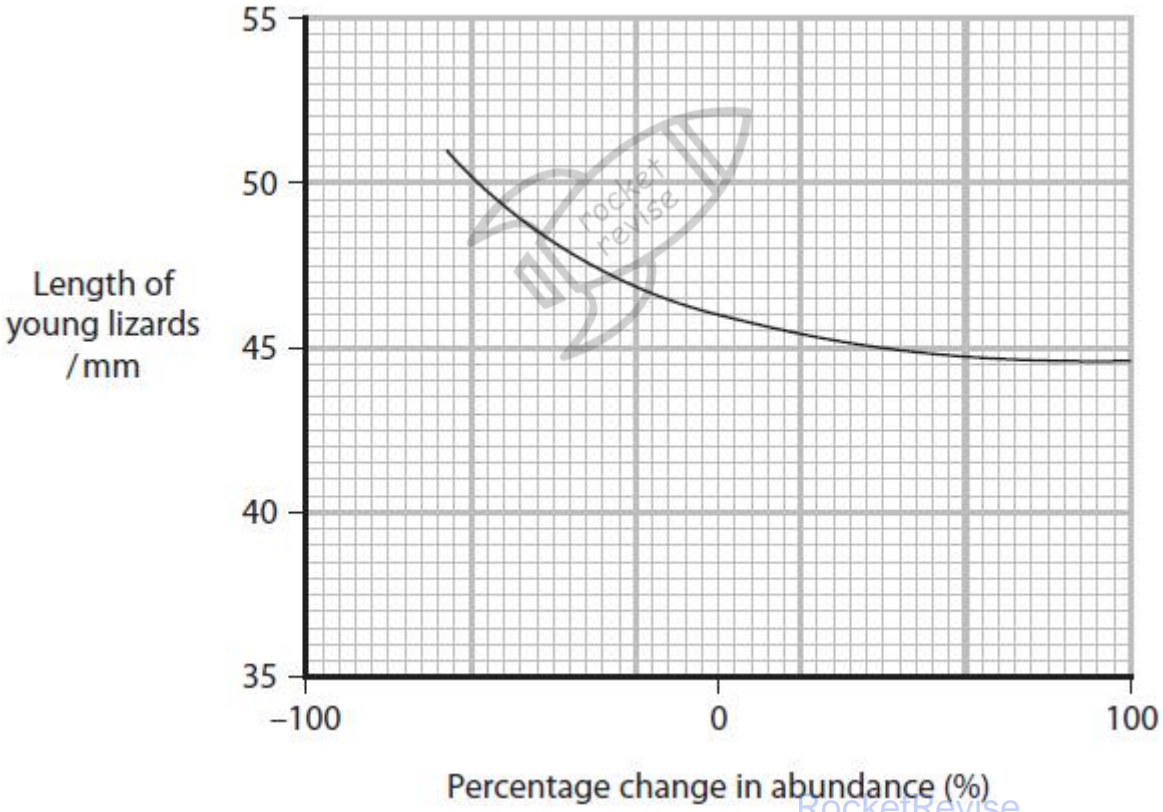
.....

.....

.....

(ii) The scientists used the data to predict a relationship between the length of the young lizards and the percentage change in abundance of lizards over the 12 year period.

The graph shows this relationship.



Describe the relationship between percentage change in abundance of lizards and length of young lizards.

(2)

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

(iii) These scientists also looked at the length of the telomeres in the cells of these young lizards.

Explain why the abundance of the longer lizards is affected by the number of cell divisions taking place as they grow.
Use all the information in the question to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 10 marks)

Q27.

Anthropogenic climate change is considered to be a result of greenhouse gas emissions.

(a) State what is meant by the term **anthropogenic climate change**.

(2)

.....

.....

.....

.....

.....

.....

(b) (i) Name **two** greenhouse gases.

(1)

.....

.....

(ii) Explain the role of greenhouse gases in climate change.

(2)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

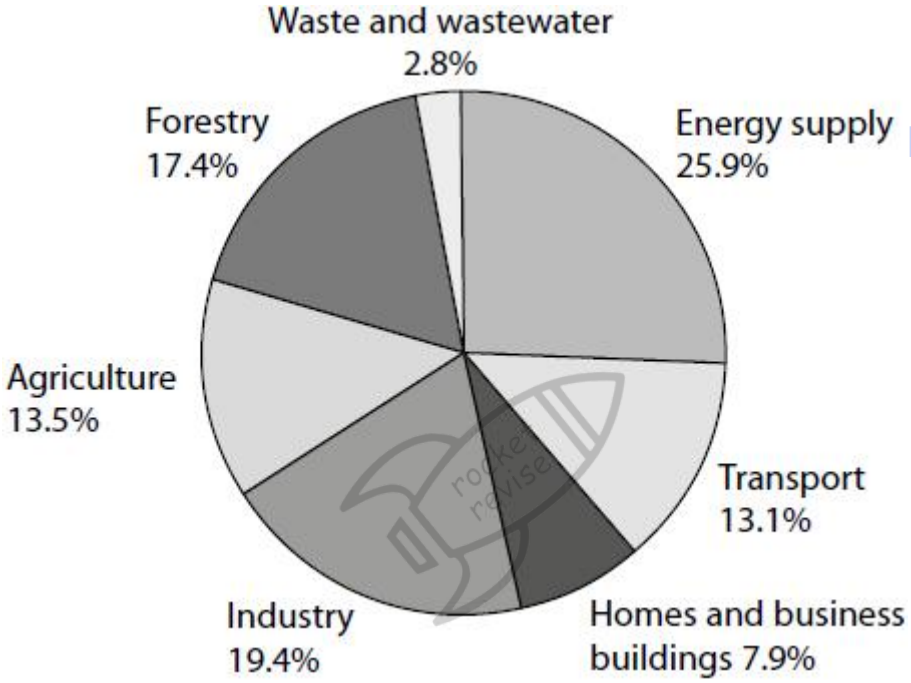
.....

.....

.....

.....

*(c) The pie chart shows the relative contribution of different sources of greenhouse gas emissions.



RocketRevise

Explain the relative contribution of each source of greenhouse gas. Use the information in the pie chart and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

(Total for question = 11 marks)

RocketRevise

Q28.

Phytoplankton are single-celled eukaryotic organisms that can photosynthesise. They live in the oceans.

A coccolithophore is a phytoplankton enclosed by plates of calcium carbonate.

The photograph shows one species of coccolithophore, *E. huxleyi*.

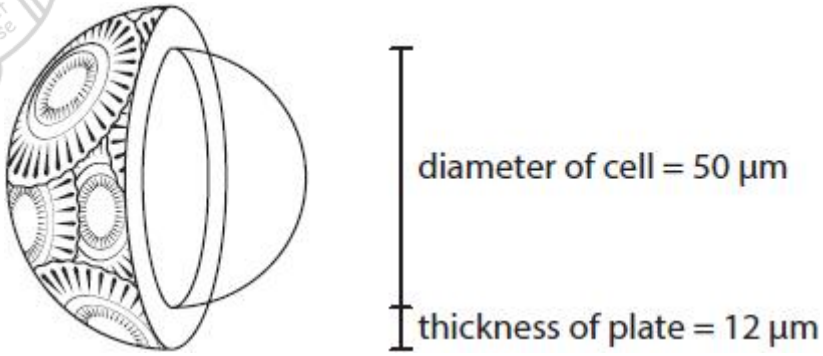


(Source: © The Natural History Museum / Alamy Stock Photo)

RocketRevise

(a) In one drop of seawater there are between 1 000 and 10 000 *E. huxleyi*.

The diagram shows the dimensions of one *E. huxleyi*.



Calculate the **total** volume of this *E. huxleyi*.
Give your answer in mm^3.

www.RocketRevise.com

www.RocketRevise.com

Use the formula:

$$V = \frac{4}{3}\pi r^3$$

(3)

Answer ..... mm^3

(b) Scientists think that *E. huxleyi* may have an impact on global warming and climate change.

E. huxleyi are continually producing the plates of calcium carbonate using the carbon dioxide they have taken in from the water.

RocketRevise

(i) One *E. huxleyi* can contain 12 plates.
It takes 75 minutes to produce one plate.
Calculate the number of **hours** it takes one *E. huxleyi* to produce 12 plates.
Assume only one plate at a time is made.

(1)

Answer ..... hours

(ii) Some of the plates produced by *E. huxleyi* are shed into the ocean water.
Scientists have discovered that plates shed into the water can attach to other *E. huxleyi*.
Describe how scientists could show that plates shed into the water can attach to **other***E. huxleyi* cells.

(3)

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

*(iii) The plates can carry viruses that infect the *E. huxleyi*.
Infected *E. huxleyi* produce a glue that sticks many cells and plates together to form a 'marine snow'.
The marine snow is very dense and sinks to the bottom of the oceans.
Explain the possible impact that infected *E. huxleyi* could have on global warming and the effects of climate change.
Use the information in the question and your own knowledge to support your answer.

(6)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 13 marks)

Q29.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Seaweeds are photosynthetic organisms. They contain chloroplast pigments, some of which are similar to land plants and some of which are different.

(a) Which process can be used to separate chloroplast pigments according to their solubility?

(1)

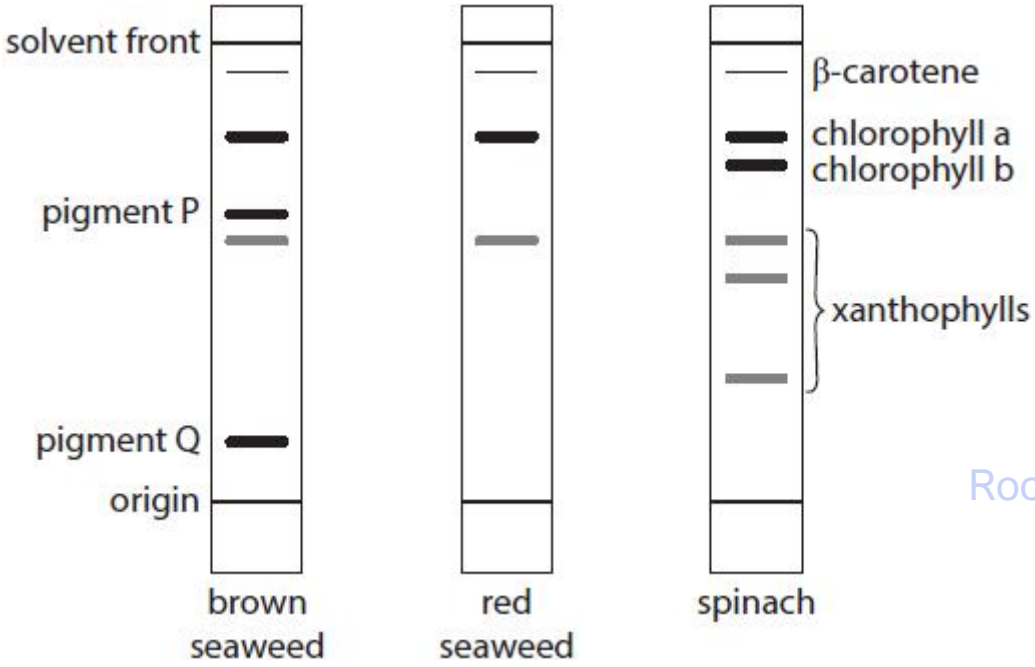
- ☒ **A** chromatography
- ☒ **B** dendrochronology
- ☒ **C** gel electrophoresis
- ☒ **D** polymerase chain reaction

(b) A student separated the chloroplast pigments found in brown seaweed, red seaweed and spinach using this method.

www.RocketRevise.com

www.RocketRevise.com

The diagram shows the results.



RocketRevise

(i) Compare and contrast the types of chloroplast pigment found in the seaweeds with those found in the spinach.
Use the information in the diagram to support your answer.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Describe how the student could identify pigment P using this method.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

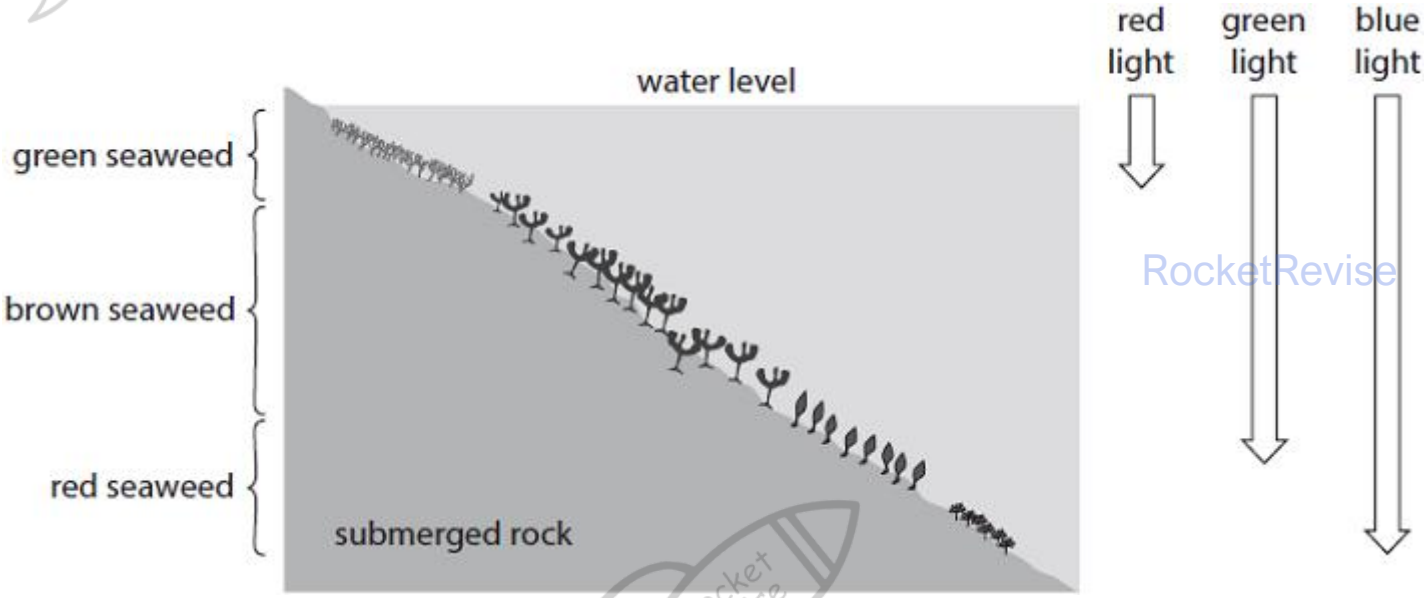
www.RocketRevise.com

.....

.....

(c) The diagram shows the position of three different types of seaweed on rock covered in water.

The diagram also shows the depth to which red light, green light and blue light penetrate into the water.



(i) Which row of the table shows the light that green seaweed can absorb?

(1)

		Green light	Red light
☒	**A**	can absorb	can absorb
☒	**B**	can absorb	cannot absorb
☒	**C**	cannot absorb	can absorb
☒	**D**	cannot absorb	cannot absorb

(ii) Explain the positions of the brown and red seaweeds on the submerged rock.

(2)

.....

.....

.....

.....

.....

.....

(Total for question = 10 marks)

www.RocketRevise.com

www.RocketRevise.com

Q30.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Apple maggot flies are thought to have evolved from hawthorn flies by sympatric speciation.

Hawthorn flies are native to North America. They lay their eggs in the berries of the hawthorn bush.

The eggs hatch into maggots and the maggots develop into adult flies.

(a) State the meaning of the term **sympatric speciation**.

(2)

.....

.....

.....

.....

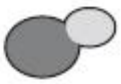
.....

.....

(b) Which row of the table represents sympatric speciation?

The shapes shaded dark grey represent the original species and the shapes shaded light grey represent the new species.

(1)

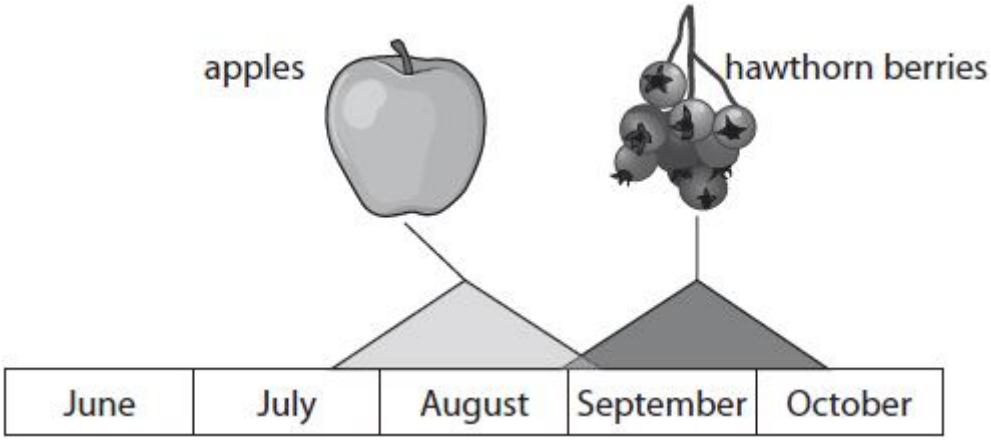
	Original species	First step in speciation	Final species
☐ A			
☐ B			
☐ C			
☐ D			

(c) Following the introduction of apple trees to North America, a small group of the hawthorn flies started laying their eggs in the apples.

The diagram shows the two types of fruit and the time of year that the fruits are produced.

www.RocketRevise.com

www.RocketRevise.com



(i) Suggest **two** adaptations needed for this change in behaviour of the flies.

(2)

RocketRevise

.....

.....

.....

.....

.....

.....

(ii) Explain what had to occur for changes in these flies to be inherited.

(3)

.....

.....

.....

.....

.....

.....

.....

RocketRevise

.....

.....

(iii) Explain why it is an advantage for the hawthorn fly and the apple maggot fly to lay their eggs in the fruits of different plants.

(2)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(iv) Suggest how these flies became reproductively isolated.
Use the information in the diagram to support your answer.

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 12 marks)

Q31.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The light-independent reactions of photosynthesis use the products of the light-dependent reactions to produce simple sugars.

(i) Which row of the table shows the products of the light-dependent reactions that are used in the light-independent reactions?

(1)

		ATP produced by	NADP
☐	**A**	cyclic photophosphorylation	oxidised
☐	**B**	cyclic photophosphorylation	reduced
☐	**C**	non-cyclic photophosphorylation	oxidised
☐	**D**	non-cyclic photophosphorylation	reduced

RocketRevise

(ii) Simple sugars have the formula $C_nH_{2n}O_n$.

Name the inorganic molecule that provides each element in a simple sugar.

(2)

C .....

H .....

O .....

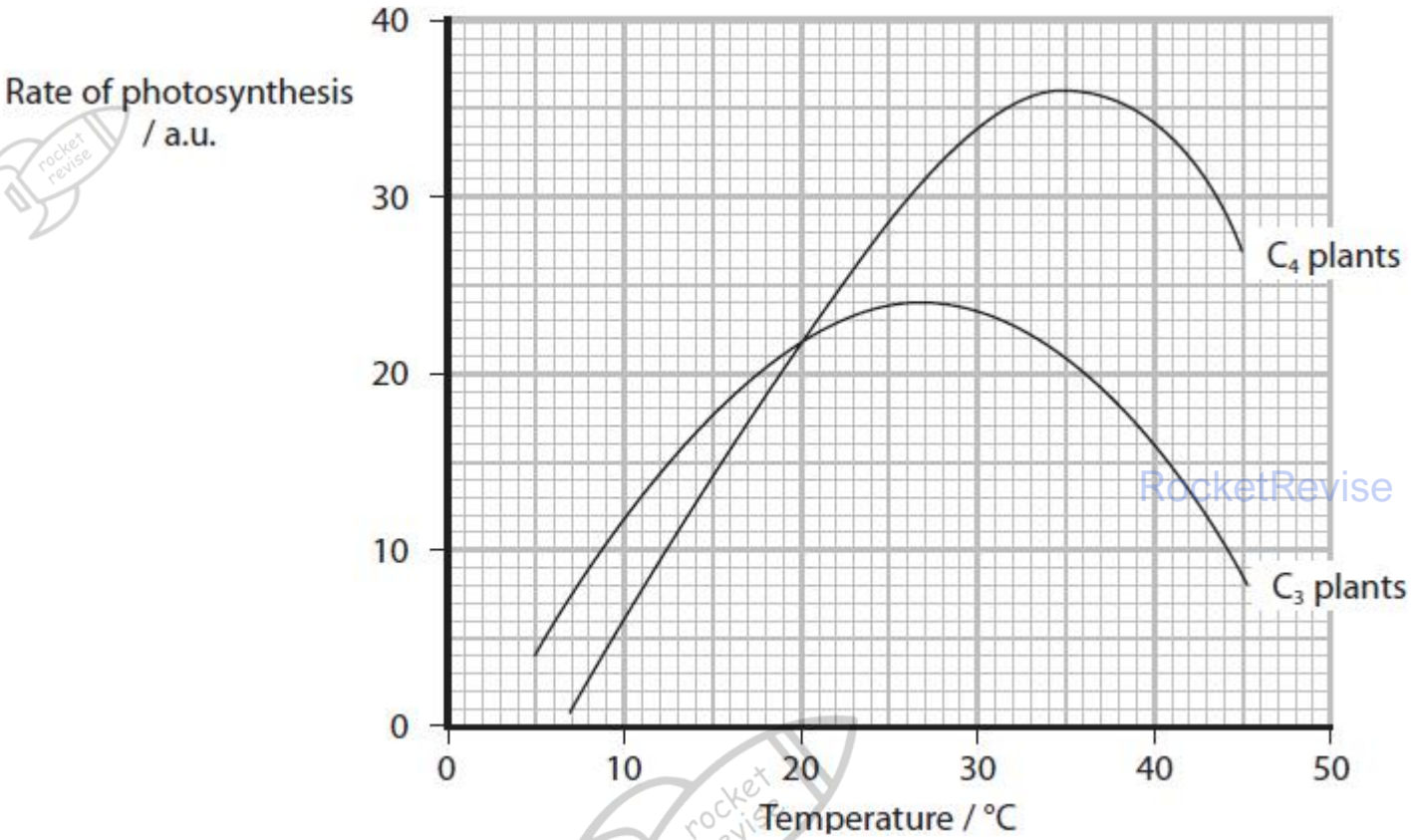
(b) Two types of plant, C_3 and C_4, have slightly different pathways of light-independent reactions.

The graph shows how the rate of photosynthesis is affected by temperature in these two types of

www.RocketRevise.com

www.RocketRevise.com

plant.



(i) Describe **three** conclusions that can be made about the differences in the effect of temperature on the rate of photosynthesis in these two types of plant.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Calculate the Q_{10} for the enzymes in C_4 plants.

Use the formula:

$$Q_{10} = \frac{R_{t+10}}{R_t}$$

where R_t is the initial rate of reaction at 10°C.

Give your answer as a whole number.

www.RocketRevise.com

www.RocketRevise.com

(2)

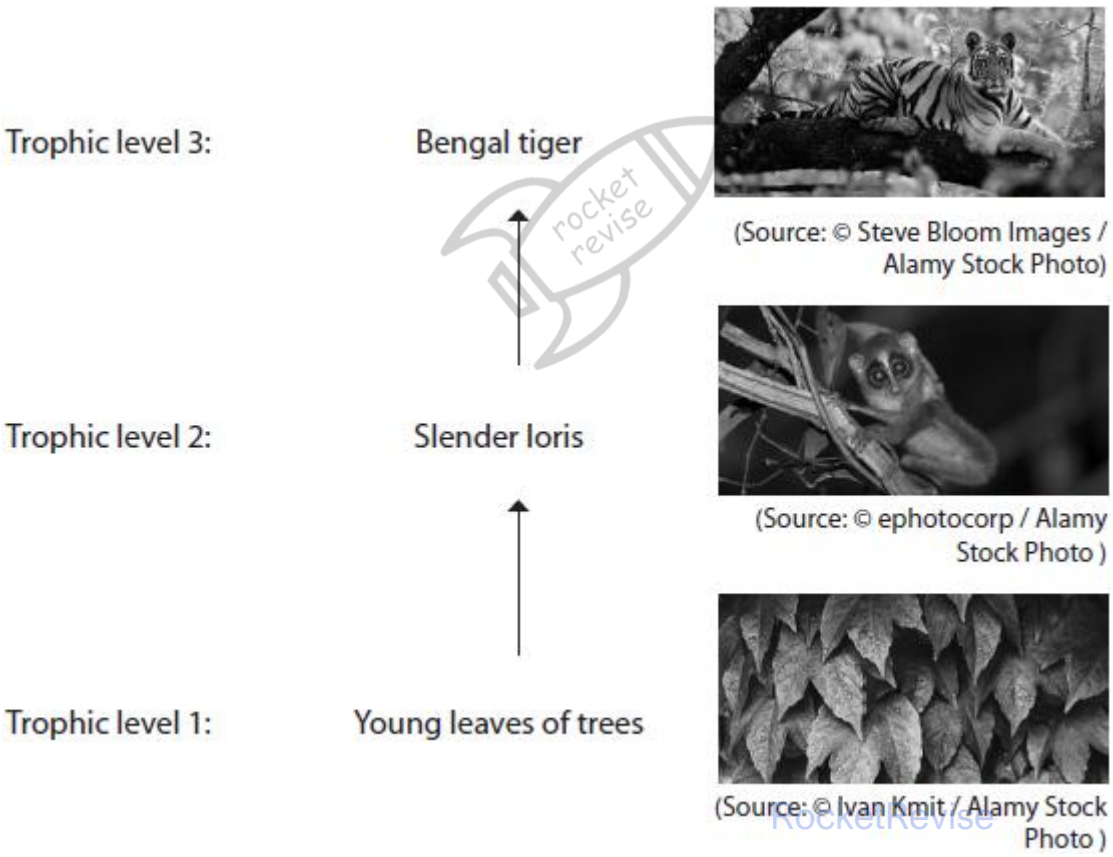
Answer .....

(Total for question = 8 marks)

Q32.

RocketRevise

The diagram shows a food chain in a southeast Asian rainforest.



(Source: © Steve Bloom Images / Alamy Stock Photo)

(Source: © ephotocorp / Alamy Stock Photo)

(Source: © Ivan Kmit / Alamy Stock Photo)

(a) Give the meaning of each of the following terms.

Give an example from this food chain in each of your answers.

(i) Habitat

(2)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(ii) Population

(2)

.....

.....

.....

.....

.....

.....

RocketRevise

(iii) Community

(2)

.....

.....

.....

.....

.....

.....

(b) Explain how the niche of the slender loris affects the distribution and abundance of the Bengal tigers.

(3)

.....

.....

.....

.....

.....

RocketRevise

.....

.....

.....

.....

(c) Southeast Asian forests are the oldest rainforests on Earth.

Explain how the age of a tree in a rainforest can be determined, without cutting it down.

(3)

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

.....

.....

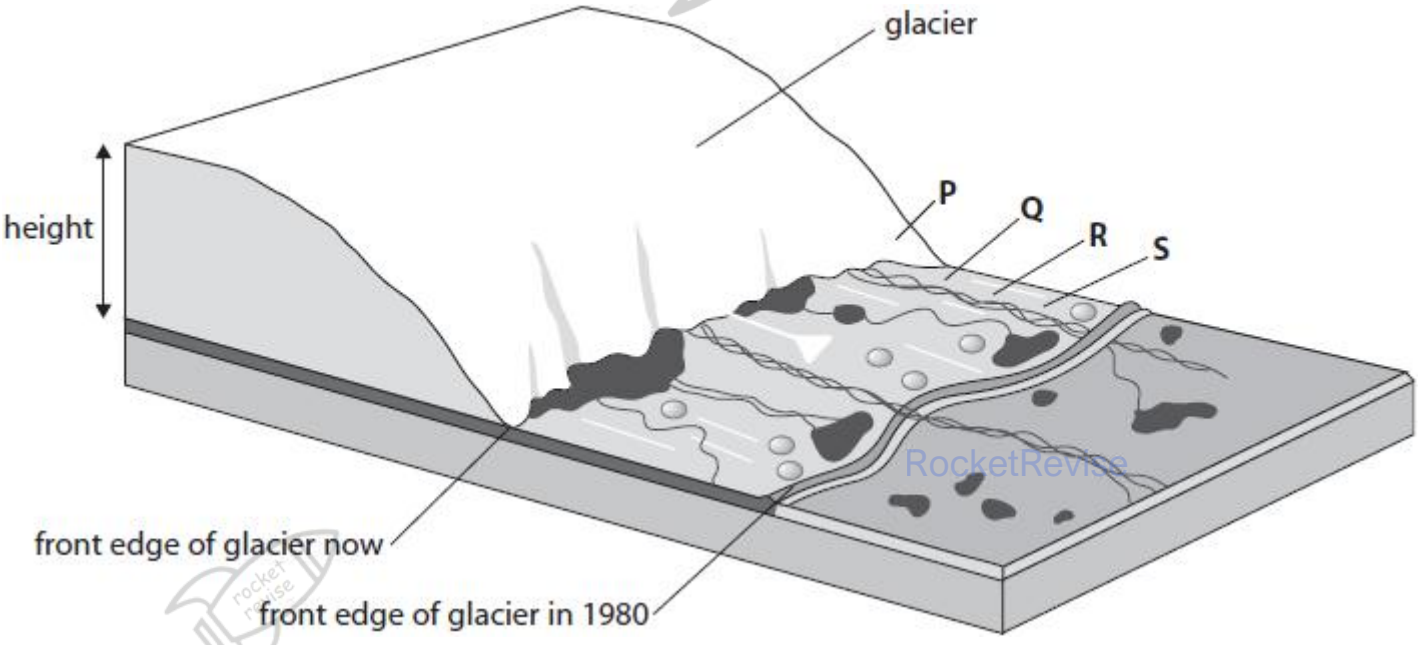
(Total for question = 12 marks)

Q33.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Glaciers are decreasing in height and retreating due to global warming.

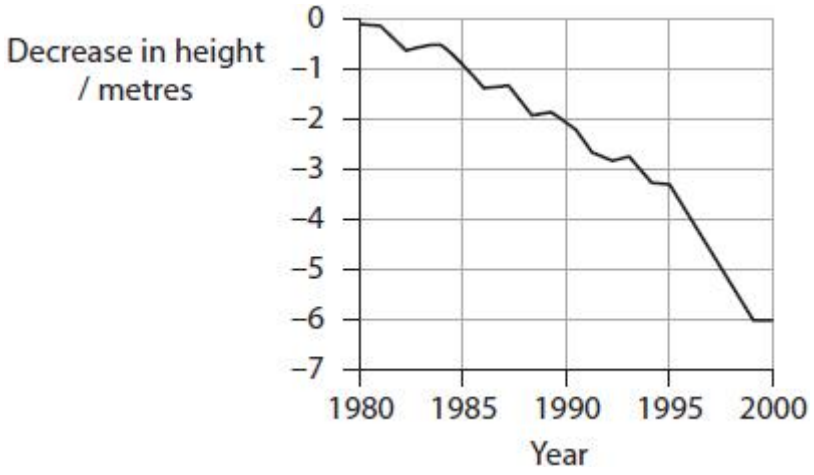
The diagram shows a retreating glacier.



(a) The graph shows the decrease in height of a glacier from 1980 to 2000.

www.RocketRevise.com

www.RocketRevise.com



(i) Calculate the mean drop in height from 1980 to 2000.
Give suitable units for your answer.

RocketRevise

(1)

Answer .....

(ii) Explain why the decrease in height is greater from 1995 to 2000 than it is from 1980 to 1985.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(b) Which letter on the diagram indicates the location of pioneer species only?

(1)

- ☒ **A** P
- ☒ **B** Q
- ☒ **C** R
- ☒ **D** S

(c) At different distances from the front edge of the glacier, there will be variations in soil conditions.

These soil conditions include: soil depth, the carbon : nitrogen ratio and the organisms present in the soil.
These differences depend on the length of time the ground has been exposed by the retreating glacier.
(i) Which term describes the changes in soil conditions and organisms with time?

www.RocketRevise.com

www.RocketRevise.com

(1)

- ☐ **A** anthropogenic
- ☐ **B** evolution
- ☐ **C** speciation
- ☐ **D** succession

*(ii) The table shows the soil carbon : nitrogen ratio in ground exposed by a retreating glacier.

Length of time that the ground has been exposed / years	Carbon : nitrogen ratio in the soil
< 200	9.8
2000	11.6
3500	10.7
7200	10.3

RocketRevise

Graph 1 shows the probability that three different types of plant will be found on ground exposed for different lengths of time.

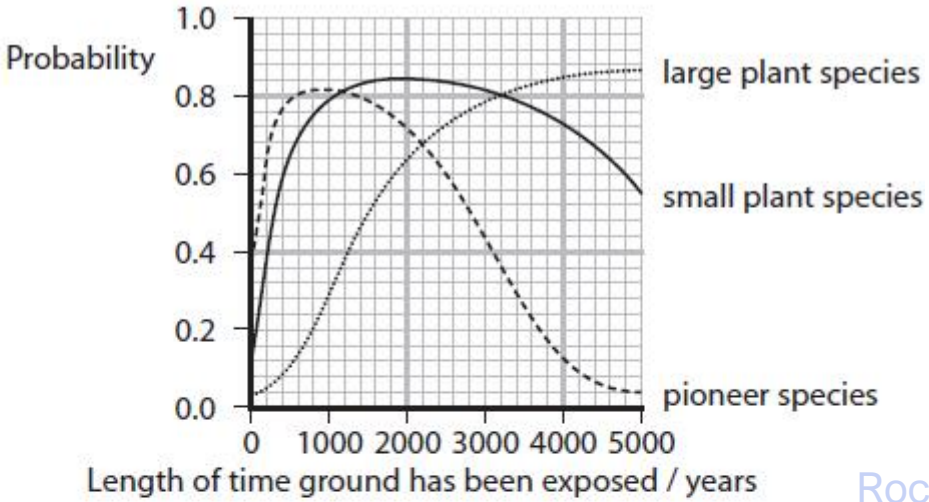
Graph 2 shows a model of plant diversity on ground exposed for different lengths of time.

RocketRevise

www.RocketRevise.com

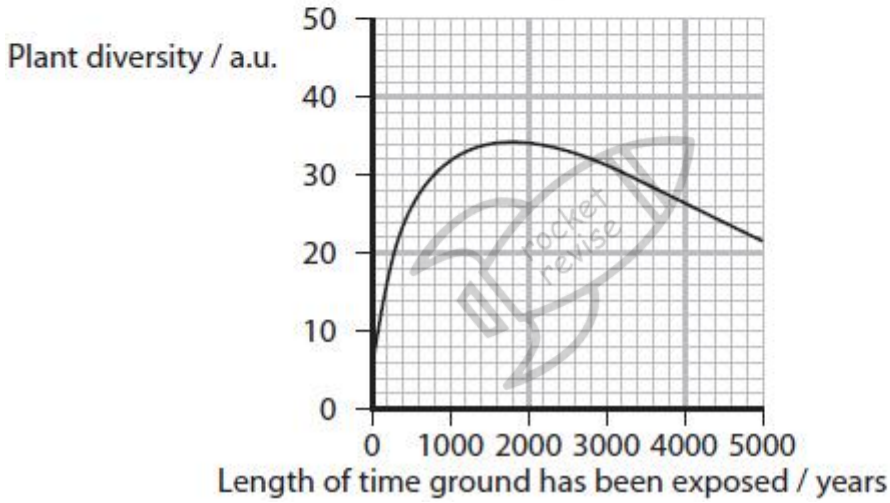
www.RocketRevise.com

Graph 1



RocketRevise

Graph 2



Explain the variations in soil conditions and the organisms found at different distances from the front edge of the glacier.

Use the information in the question and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

(Total for question = 12 marks)

www.RocketRevise.com

www.RocketRevise.com

Q34.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Textiles used in clothing can be made from:

- plants, for example cotton
- animals, for example wool and leather
- crude oil, for example nylon and polyester.

The photographs show cotton being grown and harvested.

RocketRevise



(Source: © Design Pics Inc/Alamy Stock Photo)



(Source: © Jim Parkin/Alamy Stock Photo)

In 2022, it was estimated that 92 million tonnes of textile waste was produced in the world.

Textile waste is either burned or buried underground in landfill sites.

(a) It is estimated that by 2030, there will be 134 million tonnes of textile waste produced each year.

Calculate this increase in textile waste.
Give your answer in standard form.

(1)

RocketRevise

Answer ..... tonnes

(b) The production of textiles and the removal of textile waste is an example of human activity affecting the environment.

Which term is used to describe the effect of **human** activity on the environment?

(1)

- ☒ **A** antigenic
- ☒ **B** anthropogenic
- ☒ **C** epigenetic

www.RocketRevise.com

www.RocketRevise.com

☒ **D** polygenic

(c) There are companies in Scandinavia that are producing sustainable textiles for clothing.

One company is recycling textile waste. It is removing dyes and making new fibres that can be woven. Another company is using waste from the wood industry, wheat straw and leather scraps to make fibres.

(i) Give the meaning of the term **sustainable**, as used in this context.

(1)

.....

.....

.....

*(ii) Discuss the extent to which these companies are reducing the conflict between human needs, climate change and conservation.

Use the information in the question, the photographs and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 9 marks)

www.RocketRevise.com

www.RocketRevise.com

Q35.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The Bajau people are nomadic people who free-dive in the waters around the Philippines, Malaysia and Indonesia.

They have been travelling around these waters on house boats for over 1 000 years.

They can free-dive to depths of 70 m for as long as 13 minutes, spending 60% of their working time underwater.

The photograph shows a Bajau free-diving, without breathing equipment.

RocketRevise



(Source: © Marko Reimann/Alamy Stock Photo)

Scientists measured the size of the spleens of the Bajau people and a related group of people called Saluan.

The Saluan people live on the mainland of Indonesia and do not free-dive.

The genomes of the two groups of people were analysed.

RocketRevise

(a) Animals, such as seals, that spend much of their time underwater have very large spleens.

Spleens store oxygenated red blood cells which are released into the bloodstream during a dive.
Explain why a large spleen would help the seal during its dive.

(4)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

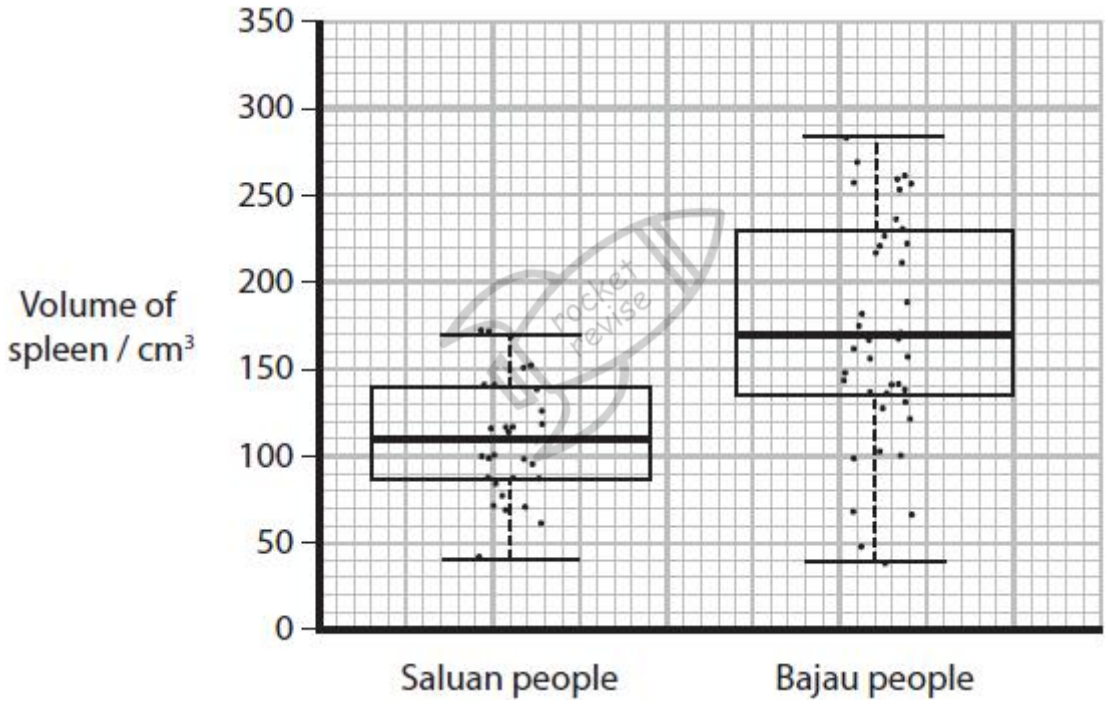
.....

.....

.....

(b) The graph shows the volumes of the spleens of individuals from the Saluan people and the Bajau people.

RocketRevise



(i) The two horizontal lines **in bold** show the middle values for each set of the volumes of spleen. Which of the following is the term that describes the middle value of a set of data?

(1)

- ☒ **A** mean
- ☒ **B** median
- ☒ **C** mode
- ☒ **D** standard deviation

RocketRevise

(ii) Describe **two** conclusions that can be made from this data.

(2)

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

*(c) The study analysed the genomes of the Saluan people and the Bajau people to identify any genetic relationships that correlated with spleen volume.

The scientists concluded that the Bajau had become genetically-adapted to diving.
Explain how the Bajau could have become genetically-adapted to diving.
Use the information in the question and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

RocketRevise

(Total for question = 13 marks)

Q36.

Very little of the light energy falling on a field of plants is absorbed by the plants and transferred to higher trophic levels.

(a) Suggest **two** reasons why very little of the light energy is absorbed by plants.

(2)

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

(b) In an American study, the light energy falling on one square metre of a field in one year was recorded as 1 700 000 kilocalories.

(i) Only one fifteenth of this light energy was converted into plant biomass.
Calculate the number of kilocalories of light energy converted into plant biomass.
Give your answer in standard form.

RocketRevise

(1)

Answer ..... kilocalories

(ii) Only 6 kilocalories became animal biomass.
Calculate the number of kilojoules (kJ) of energy that became biomass.
1 kilocalorie = 4.2 kJ.
Give your answer to **two** significant figures.

(1)

RocketRevise

Answer ..... kJ

(iii) In another field 154 000 kilocalories of energy was converted into plant biomass.
Only 25 kilocalories of this energy was passed onto the highest trophic level in a food chain.
Calculate the percentage of energy that was passed onto the highest trophic level.

(1)

www.RocketRevise.com

www.RocketRevise.com

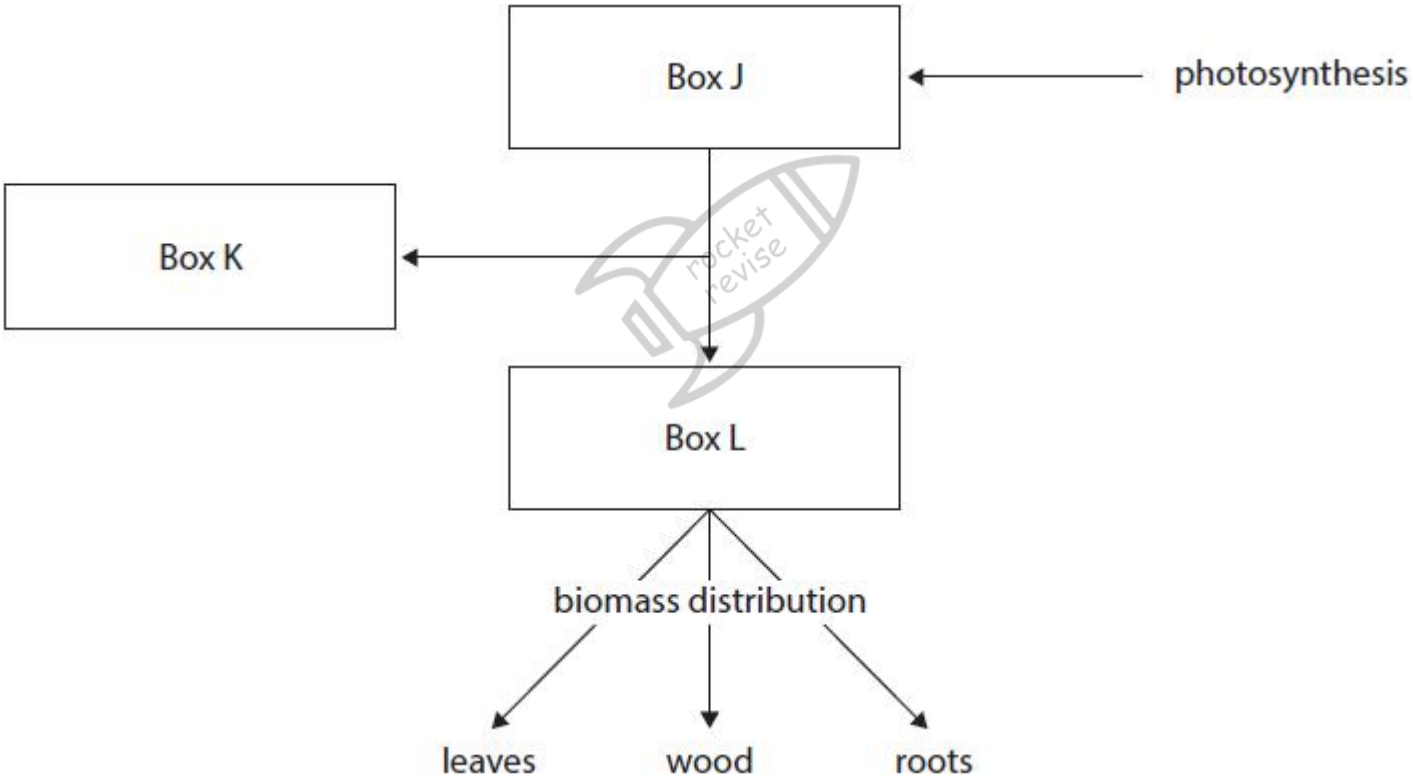
Answer ..... %

(Total for question = 5 marks)

Q37.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The diagram shows some processes linking photosynthesis to productivity and the distribution of biomass in the trees in a forest.



(a) Which row of the table identifies the box that represents net primary productivity (NPP), the box that represents gross primary productivity (GPP) and the box that represents respiration (R)?

(1)

		Box J	Box K	Box L
☒	**A**	NPP	R	GPP
☒	**B**	GPP	NPP	R
☒	**C**	GPP	R	NPP
☒	**D**	R	GPP	NPP

(b) Which are suitable units for NPP?

(1)

www.RocketRevise.com

www.RocketRevise.com

☒ **A** kJ m yr^{-1}
☒ **B** kJ m^{-1} yr^{-2}
☒ **C** kJ m^{-2} yr^{2}
☒ **D** kJ m^{-2} yr^{-1}

(c) The biomass distributed to the leaves, wood and roots of the trees in this forest was 12 500 a.u.

(i) The biomass was distributed 34% to the leaves, 37% to the roots and the rest to the wood of the trees.

Calculate how many units of biomass were distributed to the wood of these trees.

(1)

RocketRevise

Answer ..... a.u.

(ii) Which row of the table shows what can be found in the biomass of leaves and roots of trees?

(1)

		Leaves	Roots
☒	**A**	chlorophyll and starch	chlorophyll and protein
☒	**B**	chlorophyll and glycogen	lignin and protein
☒	**C**	chlorophyll and starch	starch and protein
☒	**D**	glycogen and cellulose	cellulose and protein

(d) Forests are examples of ecosystems.

State what is meant by the term **ecosystem**.

(1)

.....

.....

.....

(Total for question = 5 marks)

Q38.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

www.RocketRevise.com

www.RocketRevise.com

Redonda is a small island in the Caribbean.

It is the remains of an extinct volcano. Over thousands of years, it became home to wildlife including species of lizards only found on this island.

Following the introduction of rats and goats onto the island, the wildlife was destroyed and the island was reduced to bare rock again.

A conservation project started in 2016 to remove all the rats and goats from the island and now the biodiversity is increasing.

The photographs show some goats and some rats.



(Source: © Ian West / Alamy Stock Photo)



(Source: © Ernie Janes / Alamy Stock Photo)

(a) Which term describes the events that took place to change the extinct volcano into a habitat for many species?

(1)

- ☒ **A** entomology
- ☒ **B** evolution
- ☒ **C** speciation
- ☒ **D** succession

(b) Which term describes the species of lizards found only on Redonda?

(1)

- ☒ **A** Archaea
- ☒ **B** endemic
- ☒ **C** epigenetic
- ☒ **D** sustainable

(c) Suggest how the rats and goats destroyed the wildlife.

Use the information in the photographs to support your answer.

(2)

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(d) Describe how to determine the biodiversity of Redonda to show the effectiveness of the conservation project.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question =7 marks)

Q39.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Succession can result in the formation of different types of habitat.

(a) State the meaning of the term **succession**.

(1)

.....

.....

.....

(b) Which of the following describes a habitat?

(1)

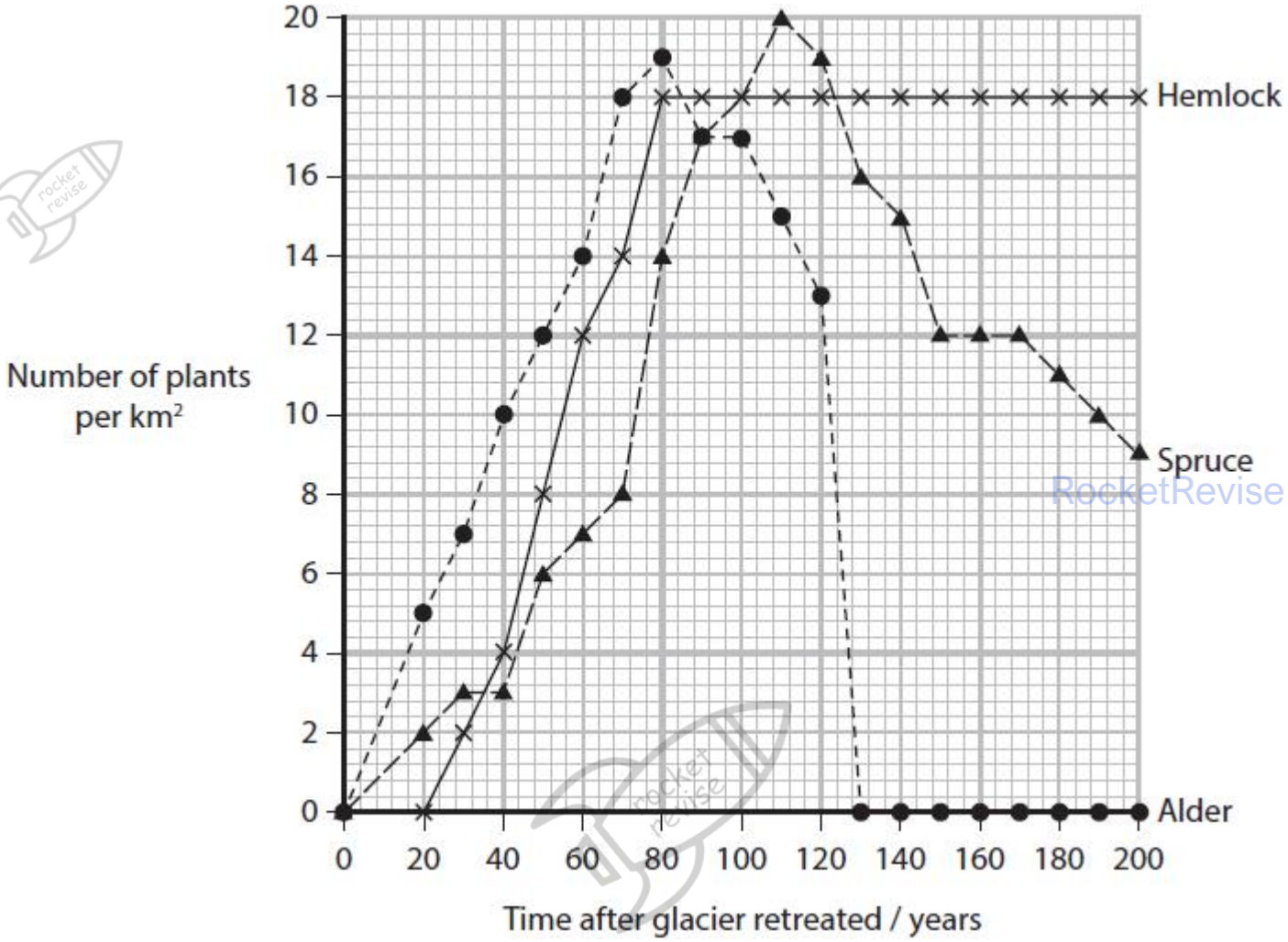
- ☒ **A** a group of interacting organisms
- ☒ **B** a place that meets the environmental conditions an organism needs to survive
- ☒ **C** the number of one species found in a particular place
- ☒ **D** the niche of an organism

*(c) Succession takes place as a glacier retreats. As the ice melts, bare rock is exposed.

The graph shows the changes in the numbers of three different types of plant, alder, spruce and hemlock, with time after a glacier has retreated.

www.RocketRevise.com

www.RocketRevise.com



Explain the changes in the numbers of the three types of plant with time after the glacier has retreated. Use the information in the question and your own knowledge to support your answer.

(6)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

.....

.....

.....

.....

(Total for question = 8 marks)

RocketRevise

Q40.

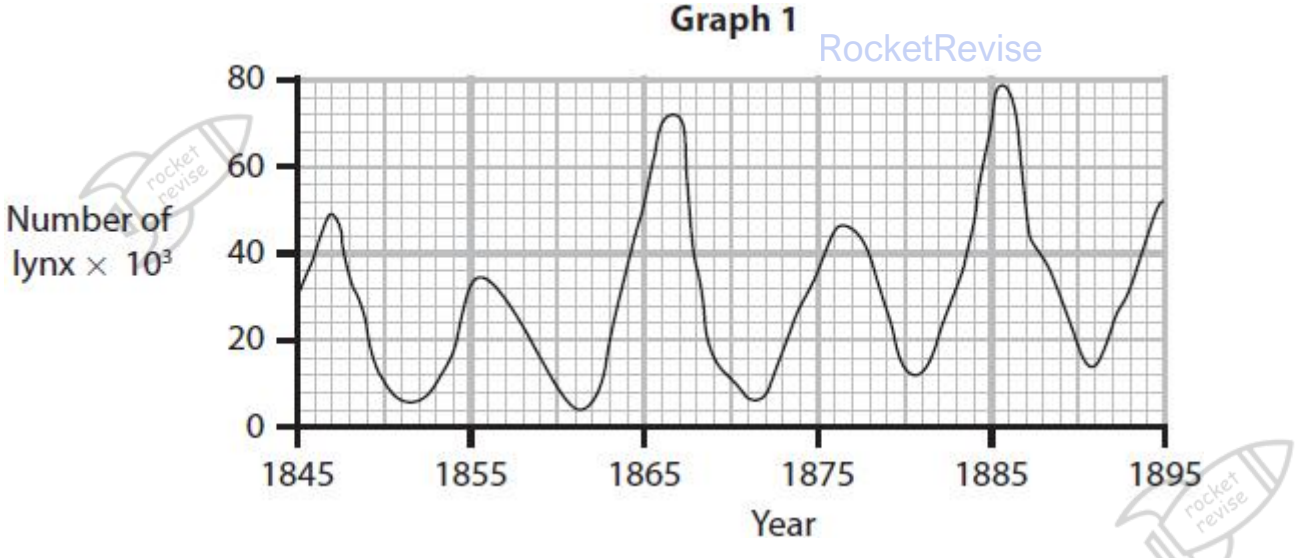
The photograph shows a Canadian lynx and a hare.



(Source: © FLPA / Alamy Stock Photo)

The lynx is a type of cat and is a predator of the hare.

Graph 1 shows the changes in the number of lynx from 1845 to 1895.



www.RocketRevise.com

www.RocketRevise.com

(a) (i) In 1863, there were 150 000 hares.

Calculate how many times more hares there were than lynx.

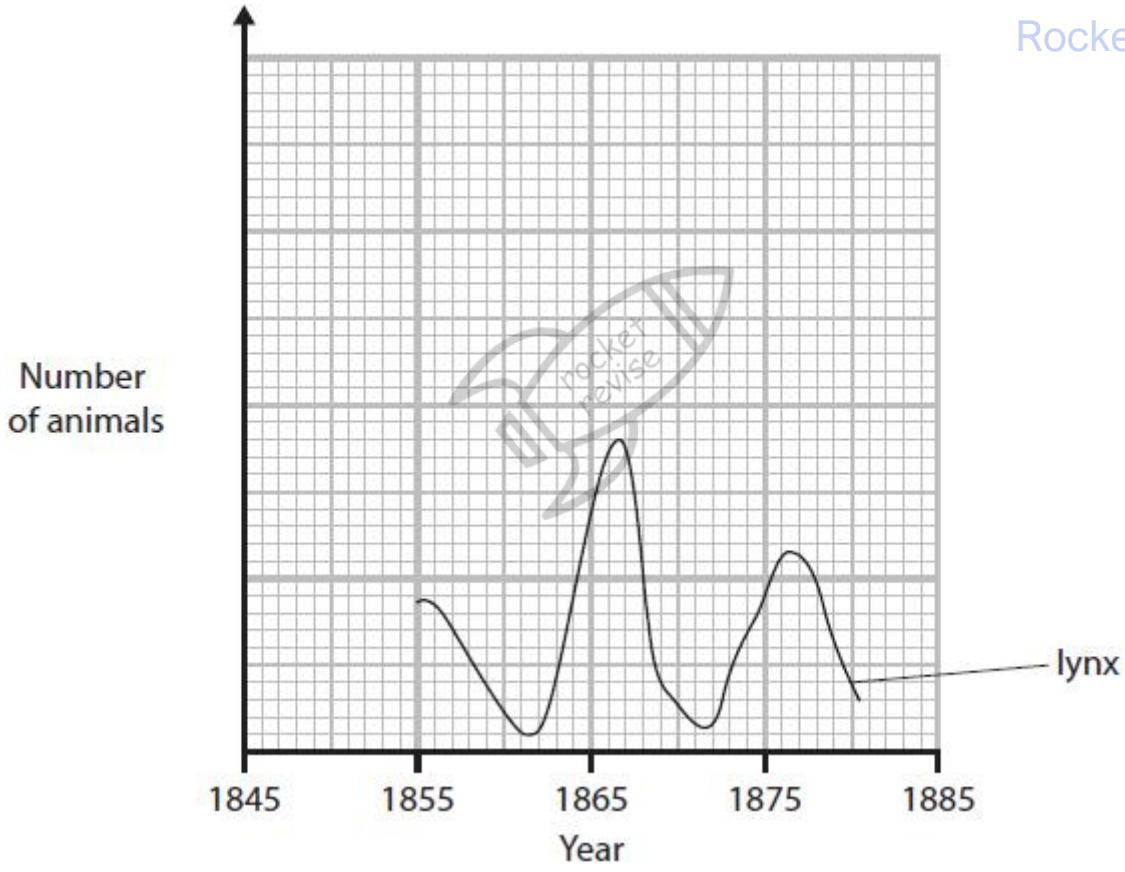
(1)

Answer .....

(ii) Graph 2 shows part of graph 1.

Sketch a line on graph 2 to show the likely changes in the numbers of hares from 1855 to 1880.

Graph 2



(2)

(iii) Explain the changes in the numbers of lynx and hares.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

.....

(b) In many ecosystems, the relationship between one predator species and one prey species is less predictable.

Suggest **two** reasons why this relationship is not always predictable.

(2)

.....

.....

.....

.....

.....

.....

(Total for question = 8 marks)

Q41.

Many reptiles lay their eggs in the sand.

The sex of these reptiles is determined by the temperature of the sand surrounding the developing eggs.

Turtles are one example of these reptiles.

The photograph shows two young turtles, one hatching from an egg.

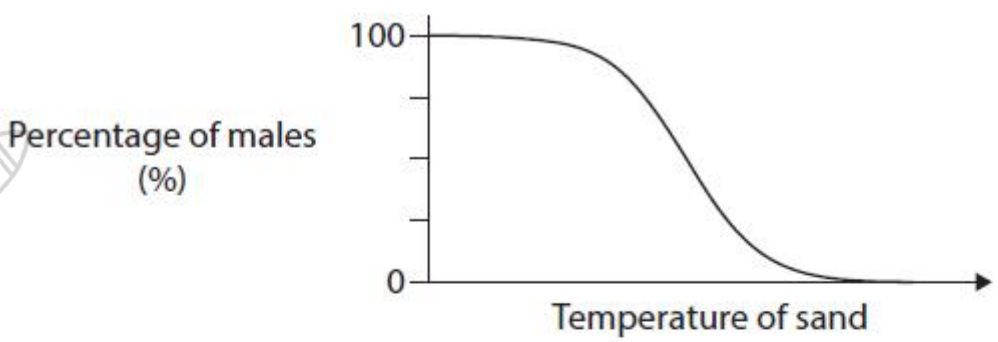


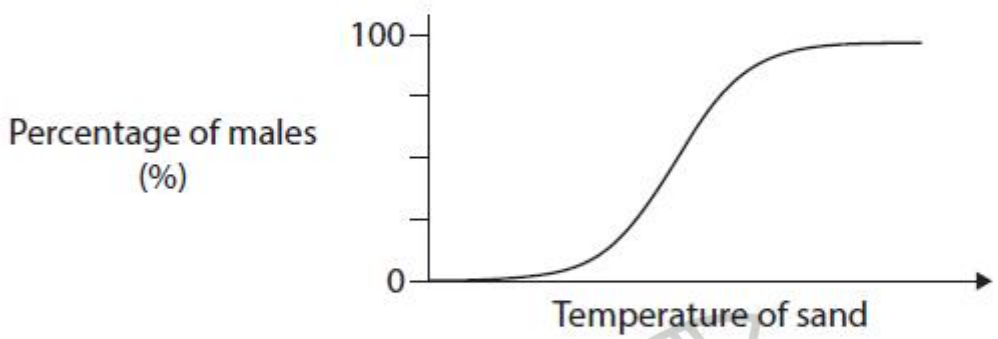
(Source: © Tierfotoagentur / Alamy Stock Photo)

(a) The graphs show the effect of temperature of the sand on the percentage of males that hatch, in three types of reptiles.

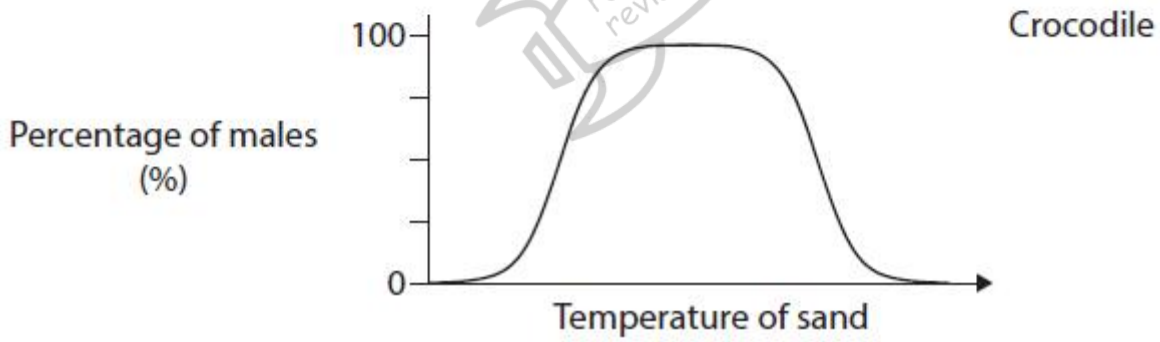
www.RocketRevise.com

www.RocketRevise.com





RocketRevise



(i) The pivotal temperature (Tpiv) is the temperature at which equal numbers of males and females are produced.
Label the Tpiv for turtles on the graph.

(1)

(ii) Describe the effect of the temperature of the sand on the percentage of males produced in these three reptiles.

RocketRevise

(3)

.....

.....

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

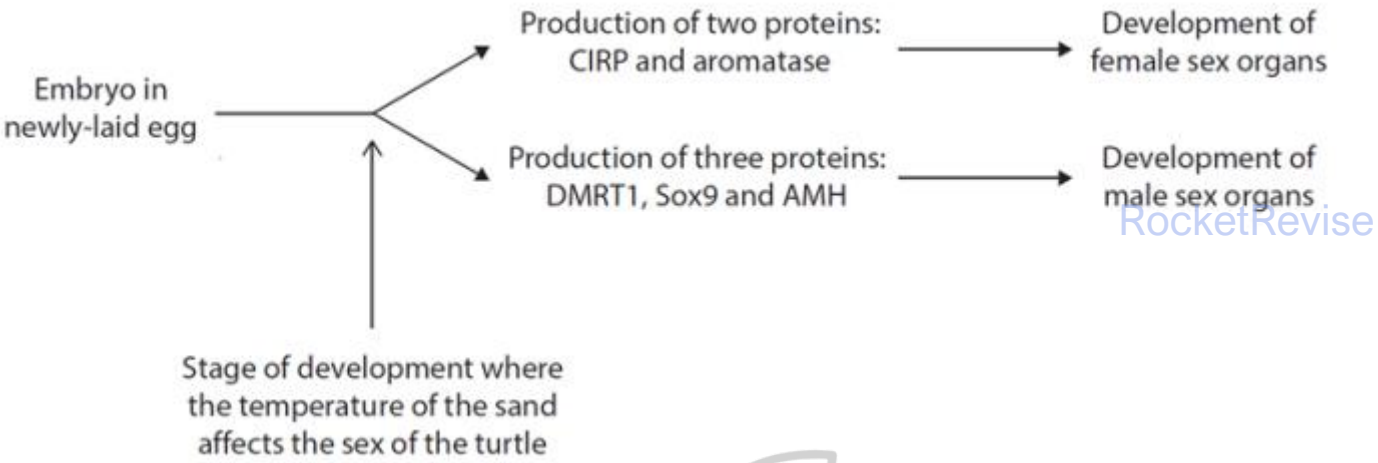
www.RocketRevise.com

.....

.....

(b) The embryos in newly-laid eggs have immature reproductive organs that can develop into either male reproductive organs or female reproductive organs.

In turtles, sex is determined by differential gene expression.
The diagram shows how the sex of a turtle is determined.



RocketRevise

Explain how temperature determines the sex of turtles.
Use the information in the graph in part (a) and the diagram to support your answer.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

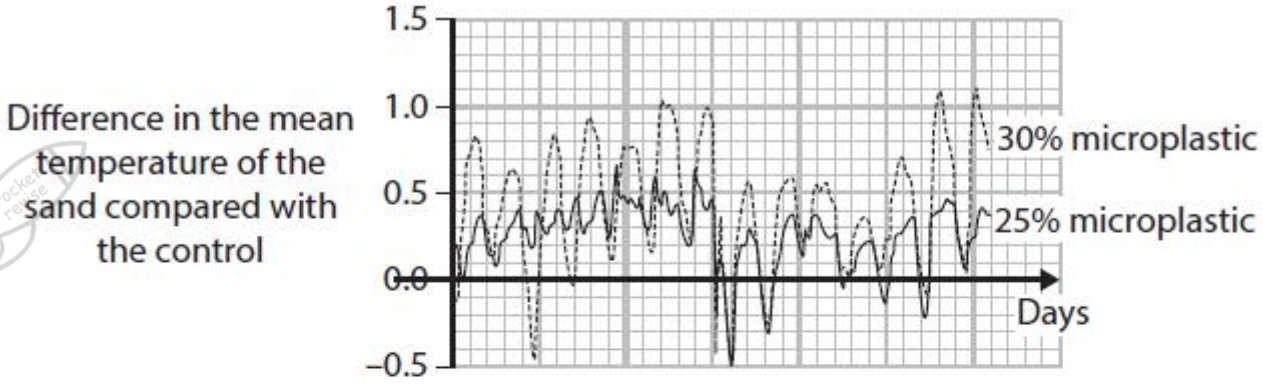
.....

(c) Anthropogenic activities such as the use of microplastic can affect the temperature of the sand.

The effect of microplastic on the temperature of the sand was investigated during the breeding time of the turtles.
Several tubs were filled with sand containing two different percentages of microplastic.
Tubs containing sand only were used as controls.
The graph shows the differences in the mean temperatures of the sand containing 25% and 30% microplastic compared with the control.
The table shows the mean temperature of the sand during this investigation.

www.RocketRevise.com

www.RocketRevise.com



Content of tubs	Mean temperature / °C	Standard deviation / °C
Control	30.047	3.088
25% microplastic	30.294	3.095
30% microplastic	30.473	3.017

(i) Name **one other** anthropogenic activity that could affect the temperature of the sand.

(1)

.....

.....

*(ii) The Tpiv for flatback turtles is 29.4 °C.
A temperature of 27.7 °C will result in all males and a temperature of 31.1 °C will result in all females being produced.
The Tpiv for green sea turtles is 30.0 °C.
Discuss whether or not scientists should be concerned that the use of microplastics could be endangering turtles.
Use all of the information in the question and your own knowledge to support your answer.

(6)

(Total for question = 15 marks)

Q42.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Tigers are on the verge of extinction, despite a recent increase in numbers.

In 2010 there were 3 200 tigers and in 2023 there were an estimated 5 574 tigers in the world.

The Bengal tiger is the most abundant with approximately 3 100 animals in India and its neighbouring countries in 2023.

www.RocketRevise.com

www.RocketRevise.com

The photograph shows a Bengal tiger.



(Source: © Zoonar GmbH / Alamy Stock Photo)

RocketRevise

(a) (i) Calculate the percentage increase in the total number of tigers in the world from 2010 to 2023.

(1)

Answer .....%

(ii) Which is the ratio of Bengal tigers to other types of tigers in the world in 2023?

(1)

- ☒ **A** 0.556 : 1
- ☒ **B** 0.798 : 1
- ☒ **C** 1.253 : 1
- ☒ **D** 1 : 1.798

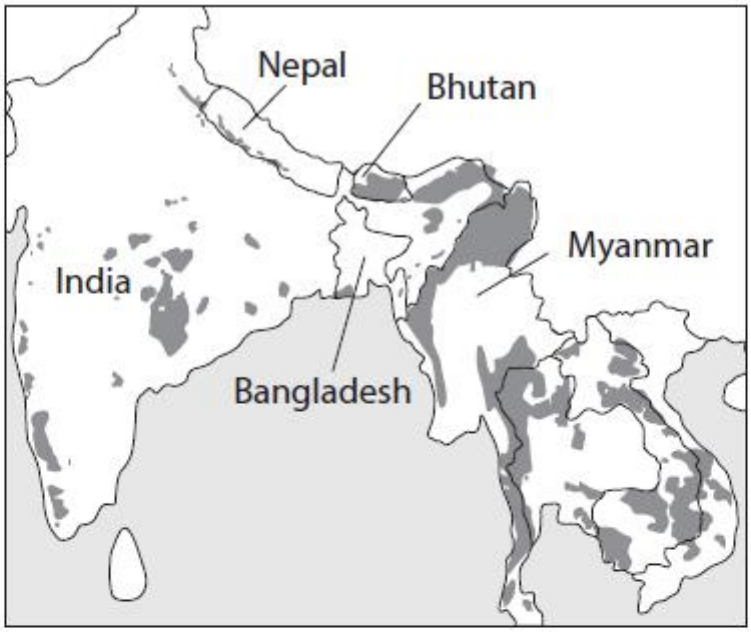
(b) The first map shows the distribution of Bengal tigers in India and its neighbouring countries.

The dark-shaded areas are where the tigers are found.

RocketRevise

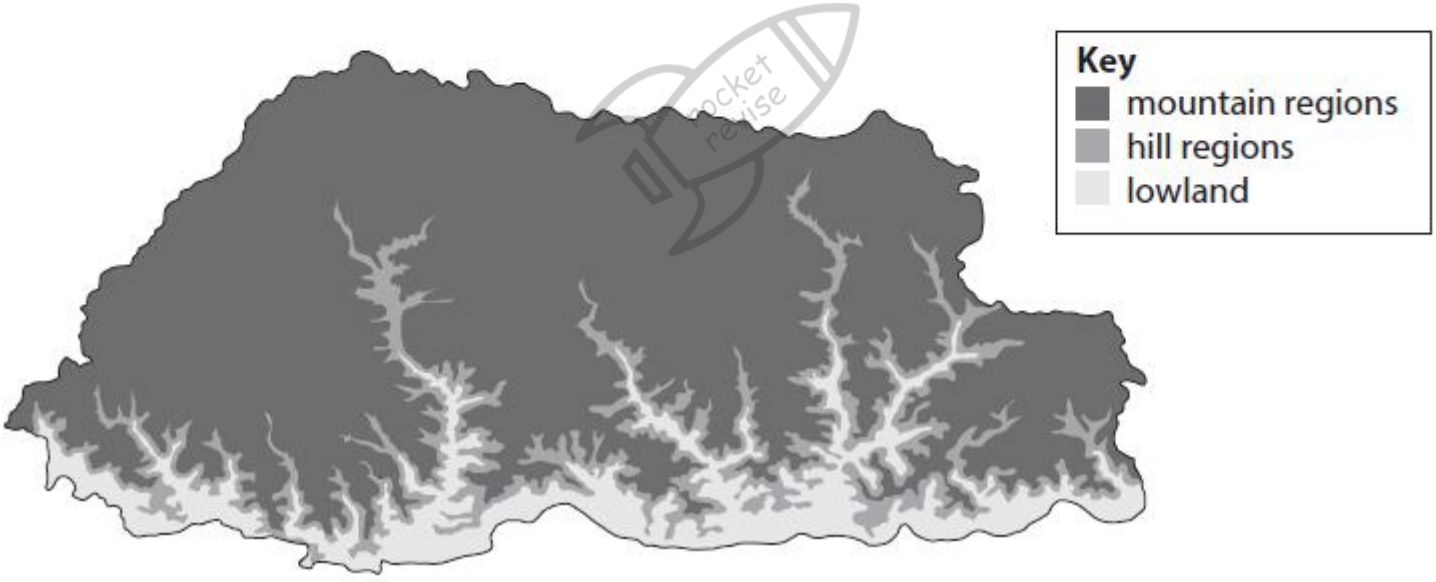
www.RocketRevise.com

www.RocketRevise.com



RocketRevise

The second map shows the mountain regions and hill regions in Bhutan.



In 2018, the Tiger Action Plan for Bhutan was started by the Royal Government of Bhutan. Conservation groups and the government were hoping to increase the number of tigers in Bhutan. This would establish Bhutan as a tiger corridor to link tiger populations in the neighbouring countries.

(i) Before the Tiger Action Plan was started, conservation groups had to prove that tigers were present in Bhutan, especially in the mountain regions shown in the second map.

Give **two** reasons why this was important.

(2)

.....

.....

.....

.....

.....

.....

www.RocketRevise.com

www.RocketRevise.com

(ii) Deduce the importance of establishing a tiger corridor in Bhutan for a greater increase in the number and diversity of tigers.
Use the information in the first map to support your answer.

(4)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(Total for question = 8 marks)

RocketRevise

RocketRevise

www.RocketRevise.com